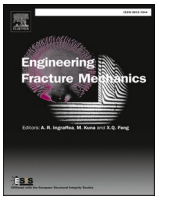
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Finite element modeling of dynamic crack branching using the moving mesh technique

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ABSTRACT

Crack branching in brittle materials is a critical phenomenon that accelerates fragmentation and may lead to sudden failure. This work presents an enhanced finite element model to simulate dynamic crack propagation and branching using a Moving Mesh (MM) technique based on the Arbitrary Lagrangian–Eulerian (ALE) formulation. The MM technique dynamically adapts the computational mesh to crack evolution without frequent remeshing, significantly improving computational efficiency. Crack initiation, propagation, and branching are governed by fracture mechanics-based criteria. Unlike traditional FEM procedures, the proposed approach avoids using specific branching angle criteria, which typically provide the angle between new branches and the original crack plane immediately after a crack branching event. Leveraging the efficacy of the MM technique, the branching angle is determined naturally, thus simplifying the modeling process and reducing the need for user-defined parameters. The accuracy and robustness of the proposed model are demonstrated through three benchmark cases, with results validated against experimental and numerical data from the literature. The simulations show that the method reliably predicts crack paths and crack branching time, highlighting its potential as a powerful tool for investigating dynamic fracture processes in brittle materials.

1. Introduction

In brittle materials, fracture mechanisms often develop dynamically because plastic dissipation is insufficient to decelerate crack propagation. When a propagating crack approaches a critical velocity, the propagation process becomes unstable due to increased roughness on the fracture surface and the formation of microscopic branches [1–4]. As the propagation velocity further increases, these micro-branches grow and coalesce into larger macro-branches, causing the primary crack to split into multiple distinct branches [5,6]. This branching phenomenon negatively affects the structural integrity of the material by accelerating fragmentation, thereby rapidly reducing its load-bearing capacity and increasing the risk of catastrophic and sudden failure.

Since brittle materials are commonly used in many engineering applications, a comprehensive understanding of the mechanics behind the crack branching phenomenon, particularly the causes of onset and potential consequences, is crucial for ensuring adequate safety levels in operational conditions [7–9].

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Nomenclature

ALE	Arbitrary Lagrangian-Eulerian.
$\mathbf{C}$	Fourth-order elastic constitutive tensor.
DSIFs	Dynamic Stress Intensity Factors.
$\boldsymbol{J}_{\boldsymbol{\Psi}}$	Jacobian matrix of the mapping function that relates the referential and material frames.
K^*	Equivalent stress intensity factor.
K_{Id}	Dynamic crack initiation toughness of the material.
K_{ID}	Dynamic crack growth toughness of the material.
K_{IA}	Crack arrest toughness of the material.
K_b	Critical stress intensity factor for the branching of the material.
LMM	Lagrangian Multipliers Method.
MM	Moving Mesh.
O_c	Origin of the crack tip coordinate systems.
R_X	Material coordinate system.
R_x	Spatial coordinate system.
R_χ	Referential coordinate system.
$\mathbf{X}$	Material coordinate.
$\dot{\boldsymbol{X}}_R$	Convective term describing the relative motion between the referential and material frames.
V_L	Limiting crack propagation velocity of the material.
$\dot{a}(t)$	Crack tip velocity.
a_{cr}	Critical velocity for branching.
c_D	Dilatational wave speed of the material.
c_R	Rayleigh wave speed of the material.
c_S	Shear wave speed of the material.
$\mathbf{f}$	Body force vector.
f_F^0	Fracture function identifying crack onset conditions.
f_F^P	Fracture function governing the crack growth process.
$f_F^{B,v}, f_F^{B,K}$	Fracture function identifying the crack branching time.
h	Solid thickness.
l_B	Length of the branching segments after crack branching.
$l_s^{curr.}$	Current length of the stretching segment.
$l_s^{init.}$	Initial length of the stretching segment.
$\Delta l_s^{\max}$	Maximum extension of the stretching segment.
m	Dimensionless shape parameter in the dynamic crack growth toughness function.
$\boldsymbol{n}_c$	Versor identifying the direction of crack propagation.
$\boldsymbol{p}$	External tractions acting on $\partial\Omega_p$.
t_b	Time of crack branching.
$\mathbf{u}$	Displacement field of the material solid.
$\overline{\boldsymbol{u}}$	Prescribed displacements on $\partial\Omega_u$.
$\widetilde{\boldsymbol{u}}$	Nodal mesh displacement function.
$\mathbf{x}$	Spatial coordinate.
x_1^c, x_2^c	Crack tip coordinate system.
$\boldsymbol{\Phi}(\chi, t)$	Mapping function that relates the referential and the spatial frames.
$\boldsymbol{\Psi}(\chi, t)$	Mapping function that relates the referential and the material frames.
Ω	Two-dimensional body domain.
$\partial\Omega$	External boundary of Ω
$\partial\Omega_c$	Internal discontinuity.
$\partial\Omega_t$	Part of $\partial\Omega$ where external tractions act.
$\partial\Omega_u$	Part of $\partial\Omega$ where displacements are imposed.
Υ	First invariant of the isochoric Green-Lagrange strain.
$\overline{\Upsilon}$	The threshold value for Υ.
Υ_{Max}	Maximum value of Υ.
α_B	Initial branching angle.
χ	Referential coordinate.
λ	Lagrangian Multiplier Method vector.
$\theta_c(t)$	Kinking angle.
ρ	Mass density.

Early studies on crack branching were experimental, performed by employing different observation and measurement techniques (*e.g.*, high-speed photomicrography [10,11], dynamic photoelasticity [12], Scanning Electron Microscopes [13], and Digital Image Correlation (DIC) [14,15]). These studies have significantly advanced knowledge regarding the mechanics of crack branching, highlighting key contributing factors such as crack propagation velocity and stress distributions near crack tips. Despite the valuable knowledge gained from the experiments, they have inherent limitations. Material defects and boundary conditions introduce variability, thus making it challenging to conduct controlled and repeatable experiments. In addition, experimental approaches are costly and often require specialized equipment, thereby limiting the scope of studies and the ability to investigate complex fracture scenarios.

During the last decades, researchers have increasingly turned to numerical methods to study the fracture behavior of materials [16–20]. Numerical approaches represent a powerful and cost-effective alternative to traditional experimental tests. These approaches allow researchers to model crack propagation mechanisms inside the material under controlled conditions, allowing for systematic variation of parameters such as material properties and loading patterns [21–23]. This capability facilitates the analysis of stress redistribution in a manner that is difficult to achieve experimentally. Several numerical approaches have been developed to simulate dynamic crack propagation mechanisms in brittle materials, including crack branching. These approaches are generally classified according to how they represent the crack in the computational domain (*i.e.*, the so-called crack model), distinguishing between discrete and smeared crack models [5].

Discrete crack models represent cracks as physical discontinuities that evolve within the computational domain. These models have been extensively implemented in numerical methods such as the Finite Element Method (FEM), the Extended Finite Element Method (XFEM), the Cracking Elements Method (CPM), and the Cracking Particles Method (CPM). Each of these numerical methods has advantages and disadvantages.

In FEM-based numerical methods, crack propagation processes are typically modeled using either remeshing techniques [24,25] or element deletion models [26,27]. Remeshing techniques update the computational mesh as the crack grows, ensuring an accurate representation of the crack path and any potential crack branching [28]. Although these techniques offer high accuracy, they are computationally expensive due to the need for continuous mesh refinements. Besides, projection solutions from older to new mesh can lead to numerical instability issues. On the other hand, element deletion models simulate crack propagation by progressively removing damaged elements based on predefined fracture criteria [29]. These methods are more computationally efficient than remeshing techniques. However, they may introduce mesh dependency issues, resulting in non-physical fracture paths unless adequately regularized. Another widely used approach in the context of FEM is the Cohesive Zone Model (CZM) [16,30,31], which represents fracture using cohesive interface elements governed by a traction–separation law. Instead of explicitly tracking cracks, this approach predefines cohesive interfaces along the boundaries of the finite elements inside the computational mesh, allowing cracks to nucleate and propagate naturally as the separation criterion is met. This modeling strategy, commonly referred to as the diffusive interface approach, offers a stable numerical framework and eliminates the need for frequent remeshing. However, its accuracy depends on mesh topology, as cracks are constrained to develop along the boundaries of finite elements, leading to mesh dependency issues and potential path bias during numerical simulations [32].

The XFEM enhances traditional FEM-based models by incorporating enrichment functions that allow the simulation of dynamic crack propagation mechanisms, regardless of the computational mesh configuration, as discontinuities due to cracks develop through finite elements [33–35]. This feature eliminates the need for remeshing actions while allowing for modeling complex and dynamic random growing cracks, including crack branching and curving, thereby making XFEM particularly suitable for large-scale crack propagation problems. However, the XFEM faces challenges in handling multiple interacting cracks and micro-branches that increase computational costs and complexity as the number of cracks grows. In particular, this method requires special enrichment techniques to reproduce crack branching [36]. Like FEM, XFEM can use cohesive zone models to simulate crack growth, although this may introduce additional computational costs.

The Cracking Element Method (CEM) presents another discrete approach within the FEM framework for simulating crack propagation [37–39]. In the CEM, cracks are represented by one or more disconnected segments within individual finite elements. This representation involves either modifying the finite element formulation or introducing duplicated nodes along the crack path within elements, effectively creating a displacement discontinuity across these segments. The method inherently allows cracks to self-propagate based on local mechanical criteria, thereby reproducing crack propagation naturally without requiring remeshing. However, accurately capturing complex crack paths, such as significant branching or curving, can be challenging since the crack trajectory remains inherently influenced by the finite element structure and how these segments connect or extend across elements.

In contrast to FEM, XFEM, and CEM, which rely on a computational mesh, the Cracking Particles Method (CPM) is a mesh-free approach that represents the material as a collection of particles. In this approach, crack propagation occurs through the breakage of particle interactions, and complex crack paths, including crack branching phenomena, are easily managed [40–44]. However, CPM can suffer from spurious crack formations if not properly regularized and may require significant computational resources due to the large number of degrees of freedom associated with particle interactions.

Unlike discrete models, smeared crack models do not explicitly represent geometric discontinuities but rather account for the presence of the crack using diffuse damage zones that evolve within the material. Among these, Phase-Field Models (PFM) have gained significant attention due to their ability to naturally capture crack initiation, propagation, and branching without requiring explicit criteria [45–48]. These models can accurately simulate complex fracture patterns by formulating crack propagation as an energy minimization problem. However, their accuracy depends heavily on the mesh resolution, as coarse discretizations may fail to reproduce fine-scale features of the fracture process. Consequently, fine meshes are necessary, thereby significantly increasing computational costs and making large-scale simulations prohibitive.

Peridynamic (PD) models [49–52] reformulate continuum mechanics using nonlocal integral equations, allowing cracks to initiate

and propagate naturally without predefined fracture paths. These models are particularly suitable for simulating complex dynamic fracture processes, including crack branching and fragmentation phenomena. However, their high computational cost and difficulty in accurately imposing boundary conditions can limit their practical applicability.

Although both discrete and smeared models have been extensively used, researchers have sought ways to overcome their respective limitations by developing hybrid and enhanced numerical methods. These approaches combine the strengths of basic numerical methods to improve computational efficiency and accuracy in simulating crack branching. Among these, a hybrid approach has been developed by combining XFEM with the phantom node method [53], which enhances the handling of branching and intersecting cracks. By locally introducing additional degrees of freedom, this technique enhances computational stability while maintaining the flexibility of XFEM. Another notable advancement is the integration of the Phase-Field Model (PFM) with the Cohesive Zone Model (CZM) [54,55]. By introducing a regularized cohesive formulation, these hybrid approaches offer improved accuracy in crack branching predictions and reduced sensitivity to mesh discretization.

Despite these advancements, only a few hybrid numerical approaches have been developed to address the inherent limitations of standard FEM procedures [56]. FEM remains the most commonly used numerical tool in engineering due to its simplicity, versatility, and ease of implementation. However, applying traditional FEM to simulate dynamic crack propagation problems poses significant challenges. Accurately tracking the evolution of crack paths, especially during branching, often necessitates frequent remeshing of the computational domain and sophisticated crack-tracking strategies. These aspects inevitably increase computational costs and may lead to numerical instabilities or reduce the accuracy of the solution. These issues are particularly critical in dynamic scenarios, where preserving mesh quality and numerical robustness is essential.

Intending to improve the performance of standard FEM procedures in the analysis of fracture mechanics problems, some of the authors of the present work have developed an advanced FE modeling approach that employs the Moving Mesh (MM) technique based on the Arbitrary Lagrangian-Eulerian (ALE) formulation to efficiently manage the geometry evolution of the computational domain because of the growth of arbitrary cracks [57–59].

Unlike traditional FE methods, which update the computational mesh for each crack growth increment by performing a remeshing action, this strategy dynamically moves the mesh nodes around the crack tip to follow the crack propagation process. In this context, the ALE formulation ensures that the movement of the mesh nodes occurs smoothly, minimizing distortions within the finite elements of the computational mesh and thus limiting remeshing to situations where element regularity is severely compromised. This strategy has been successfully applied in various contexts, demonstrating its reliability and efficiency in simulating dynamic crack propagation phenomena in homogeneous materials [59]. However, the model presented in [59] cannot reproduce crack branching phenomena, as it lacks specific fracture criteria and implementation details necessary to model the splitting of a single propagating crack into multiple branches.

Therefore, this paper aims to enhance the numerical approach initially presented by some of the present authors in [59], expanding its capability to simulate dynamic crack propagation phenomena in brittle materials, including the potential evolution into crack branching events. Although the method has previously been validated for single-crack propagation under dynamic conditions, additional refinements are required for modeling crack branching. Key innovations include the introduction of explicit branching criteria to determine the onset of branching and the development of a refined modeling technique to accurately simulate the splitting of a single propagating crack into multiple paths. In particular, the developed modeling strategy, which leverages the flexibility of the MM technique, eliminates the need for additional criteria to establish branching angles immediately after a crack branching event, thereby facilitating the numerical simulation.

The reliability and efficiency of the proposed method have been assessed through numerical simulations concerning three relevant benchmark cases widely adopted in the literature for evaluating numerical approaches to dynamic fracture modeling.

The rest of the paper is organized as follows: Section 2 provides the theoretical background at the base of the proposed approach. In particular, the criteria adopted to capture crack branching are explained. Section 3 treats the numerical implementation, particularly describing how the splitting of a single propagation crack is modeled. Section 4 presents the numerical results, discussing the accuracy of the proposed method in capturing crack branching times and final crack paths. Finally, Section 5 summarizes the main findings and suggests possible future developments.

2. Theoretical formulation

2.1. Basics of the Moving mesh technique based on the Arbitrary Lagrangian-Eulerian (ALE) formulation.

The simulation of dynamic crack propagation phenomena in brittle materials requires adapting the computational mesh to the geometric changes induced by growing cracks, thus posing severe challenges for numerical simulations. Because cracks can propagate arbitrarily, traditional FEM-based models typically regenerate the computational mesh at every step of crack growth by performing a remeshing action to correctly represent newly formed surfaces and refine the mesh near the crack tip. This approach effectively tracks changes in the domain and ensures high-resolution discretization where stress and strain gradients are most intense. However, continuous remeshings involve significant computational burdens due to repeated mesh generation and the projection of solution fields onto the new mesh.

To reduce the remeshing actions, a possible alternative consists of the use of the Moving Mesh (MM) technique based on the Arbitrary Lagrangian-Eulerian (ALE) formulation [60–63]. Basically, the MM technique dynamically modifies the configuration of the computational mesh by repositioning its nodes according to conditions that may be part of or independent of the investigated problem. In this framework, the ALE formulation ensures smooth mesh node movements, limiting excessive distortions among finite elements of

the computational mesh. By reducing the need for continuous mesh regeneration, the MM technique based on the ALE formulation lowers computational costs while mitigating the numerical instabilities often associated with abrupt changes in the simulation domain. This capability has encouraged the use of the MM technique to simulate crack propagation processes within brittle materials. More precisely, this technique can be efficiently used to dynamically adapt the computational mesh to follow the evolution of the geometry due to arbitrarily growing cracks (see Fig. 1-a).

Typically, the ALE formulation requires that the governing equations of a continuum mechanics problem must be formulated neither to traditional material (R_X) nor spatial (R_x) coordinate systems but rather to a third frame, commonly referred to as the Referential coordinate system (R_χ), where the referential coordinate $\boldsymbol{\chi}$ identifies the nodes of the computational mesh.

Bijective mapping functions relate the referential frame R_χ to spatial and material frames (Fig. 1-b):

$$\begin{aligned} &\boldsymbol{\Phi}: R_\chi \rightarrow R_x \big| \quad \boldsymbol{\chi}(t) \rightarrow \boldsymbol{\Phi}(\boldsymbol{\chi}, t) = \boldsymbol{x}(t); \\ &\boldsymbol{\Psi}: R_\chi \rightarrow R_X \big| \quad \boldsymbol{\chi}(t) \rightarrow \boldsymbol{\Psi}(\boldsymbol{\chi}, t) = \boldsymbol{X}(t); \end{aligned} \tag{1}$$

In particular, $\boldsymbol{\Phi}(\boldsymbol{\chi}, t)$ and $\boldsymbol{\Psi}(\boldsymbol{\chi}, t)$ maps the referential frame onto the spatial and material configurations, respectively. These mapping functions enable the transformation of governing equations initially formulated in spatial or material frames into the referential frame.

For the purposes of the present study, it is necessary to achieve the ALE formulation of the governing equations of solid mechanics, conventionally formulated in the material frame (R_X). In particular, such equations contain gradient operators and time derivatives that must be consistently mapped into the referential frame. Considering a generic vector field $\boldsymbol{\nu}(\boldsymbol{X}(t), t)$ in the material frame. Using the mapping function $\boldsymbol{\Psi}(\boldsymbol{\chi}, t)$, the following equivalence occurs:

$$\boldsymbol{\nu}(\boldsymbol{X}(t), t) = \boldsymbol{\nu}(\boldsymbol{\Psi}(\boldsymbol{\chi}, t), t) \tag{2}$$

The gradient of $\boldsymbol{\nu}(\boldsymbol{\Psi}(\boldsymbol{\chi}, t), t)$ can be computed as:

$$\frac{\partial \boldsymbol{\nu}(\boldsymbol{\Psi}(\boldsymbol{\chi}, t), t)}{\partial \boldsymbol{\chi}} = \frac{\partial \boldsymbol{\nu}(\boldsymbol{\Psi}(\boldsymbol{\chi}, t), t)}{\partial \boldsymbol{\Psi}(\boldsymbol{\chi})} \frac{\partial \boldsymbol{\Psi}(\boldsymbol{\chi}, t)}{\partial \boldsymbol{\chi}} = \frac{\partial \boldsymbol{\nu}(\boldsymbol{X}(t), t)}{\partial \boldsymbol{X}} \frac{\partial \boldsymbol{\Psi}(\boldsymbol{\chi}, t)}{\partial \boldsymbol{\chi}} \tag{3}$$

By denoting with $\nabla_M(\cdot) = \partial(\cdot)/\partial \boldsymbol{X}$ and $\nabla_R(\cdot) = \partial(\cdot)/\partial \boldsymbol{\chi}$, Eq. (3) assumes the following form:

$$\nabla_R \boldsymbol{\nu} = \nabla_M \boldsymbol{\nu} \frac{\partial \boldsymbol{\Psi}(\boldsymbol{\chi}, t)}{\partial \boldsymbol{\chi}} \tag{4}$$

The last term on the right-hand side of Eq. (4) represents the Jacobian matrix of the transformation between the referential and material frames:

$$\boldsymbol{J}_{\boldsymbol{\Psi}} = \frac{\partial \boldsymbol{\Psi}(\boldsymbol{\chi}, t)}{\partial \boldsymbol{\chi}} \quad \text{with } \det|\boldsymbol{J}_{\boldsymbol{\Psi}}| \geq 0 \tag{5}$$

Noting that $\det|\boldsymbol{J}_{\boldsymbol{\Psi}}| \geq 0$ ensures the one-to-one mapping between each material and its referential coordinate at each time instant.

Starting from Eqs.(4–5), the equation that maps the material gradient of a vectorial field into the referential frame can be expressed as follows:

$$\nabla_M \boldsymbol{\nu} = \nabla_R \boldsymbol{\nu} \boldsymbol{J}_{\boldsymbol{\Psi}}^{-1} \tag{6}$$

To derive the transformation rule relating the time derivatives in the material and referential frames, consider the first-time derivative

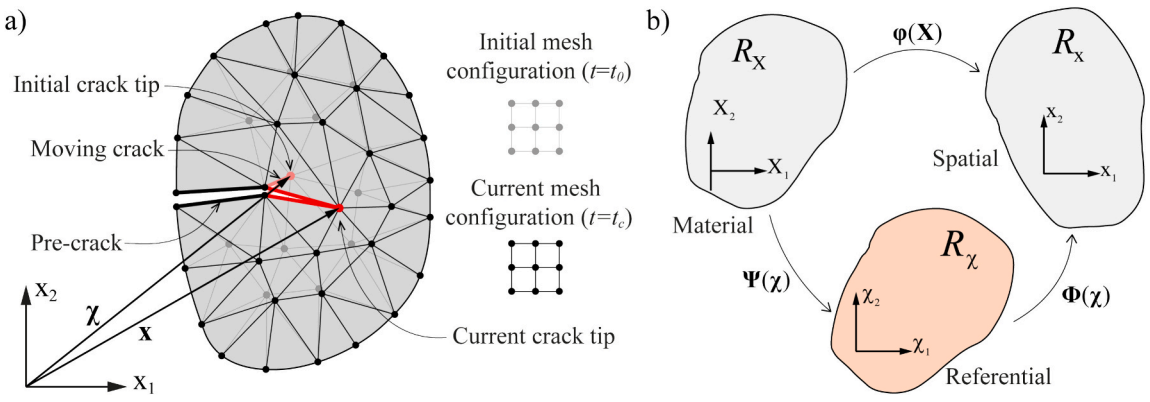


Fig. 1. The Moving Mesh technique based on the Arbitrary Lagrangian–Eulerian formulation. (a) Movement of the computational mesh nodes to simulate the dynamic growth of a crack tip that passes from an initial configuration ($t = t_0$) to a current configuration ($t = t_c$); (b) A depiction of the mapping functions that link the spatial (R_x), material (R_X), and referential (R_χ) coordinate systems.

of the vectorial field $\boldsymbol{v}(\boldsymbol{\Psi}(\boldsymbol{\chi},t),t)$:

$$\frac{d}{dt}\boldsymbol{v}(\boldsymbol{X}(t),t)=\frac{d}{dt}(\boldsymbol{\Psi}(\boldsymbol{\chi},t),t) \tag{7}$$

The referential first-time derivative is related to the material one by the following equation:

$$\frac{d}{dt}(\boldsymbol{\Psi}(\boldsymbol{\chi},t),\mathrm{t})=\frac{\partial}{\partial t}\boldsymbol{v}(\boldsymbol{\Psi}(\boldsymbol{\chi},t),t)\Big|_{\chi}=\frac{\partial}{\partial t}\boldsymbol{v}(\boldsymbol{\Psi}(\boldsymbol{\chi},t),t)\Big|_{\boldsymbol{\Psi}(\chi)}+\frac{\partial \boldsymbol{v}(\boldsymbol{\Psi}(\boldsymbol{\chi},t),t)}{\partial \boldsymbol{\Psi}(\boldsymbol{\chi},t)}\frac{\partial \boldsymbol{\Psi}(\boldsymbol{\chi},t)}{\partial t} \tag{8}$$

$$\frac{\partial}{\partial t}\boldsymbol{v}(\boldsymbol{\Psi}(\boldsymbol{\chi},t),t)\Big|_{\chi}=\frac{\partial}{\partial t}\boldsymbol{v}(\boldsymbol{X}(t),t)\Big|_{\boldsymbol{X}}+\frac{\partial \boldsymbol{v}(\boldsymbol{X}(t),t)}{\partial \boldsymbol{X}}\frac{\partial \boldsymbol{\Psi}(\boldsymbol{\chi},t)}{\partial t}\Big|_{\chi} \tag{9}$$

By denoting the first-time referential and material time derivative respectively as:

$$\dot{\boldsymbol{v}}_R=\frac{\partial}{\partial t}(\bullet,t)\Big|_{\chi}\qquad \dot{\boldsymbol{v}}=\frac{\partial}{\partial t}(\bullet,t)\Big|_{\boldsymbol{X}} \tag{10}$$

Eq. (9) assumes the following form:

$$\dot{\boldsymbol{v}}_R=\dot{\boldsymbol{v}}+\nabla_M\boldsymbol{v}\dot{\boldsymbol{X}}_R \tag{11}$$

Here, $\dot{\boldsymbol{X}}_R$ represents a convective term describing the relative motion between the referential and material frames. By taking into account Eq. (6), the first-time material derivative can be expressed in the referential frame as follows:

$$\dot{\boldsymbol{v}}=\dot{\boldsymbol{v}}_R-\nabla_M\boldsymbol{v}\dot{\boldsymbol{X}}_R=\dot{\boldsymbol{v}}_R-\nabla_R\boldsymbol{v}\boldsymbol{J}_{\boldsymbol{\Psi}}^{-1}\dot{\boldsymbol{X}}_R \tag{12}$$

Applying Eq. (12) recursively yields the second-time derivative of $\boldsymbol{v}(\boldsymbol{X},t)$ mapped into the referential frame:

$$\ddot{\boldsymbol{v}}=\ddot{\boldsymbol{v}}_R-2\nabla_R\dot{\boldsymbol{v}}_R\boldsymbol{J}_{\boldsymbol{\Psi}}^{-1}\dot{\boldsymbol{X}}_R-\nabla_R\boldsymbol{v}\boldsymbol{J}_{\boldsymbol{\Psi}}^{-1}\ddot{\boldsymbol{X}}_R+\nabla_R\left[\nabla_R\boldsymbol{v}\boldsymbol{J}_{\boldsymbol{\Psi}}^{-1}\right]\boldsymbol{J}_{\boldsymbol{\Psi}}^{-1}\dot{\boldsymbol{X}}_R\dot{\boldsymbol{X}}_R+\nabla_R\boldsymbol{v}\boldsymbol{J}_{\boldsymbol{\Psi}}^{-1}\nabla_R\dot{\boldsymbol{X}}_R\boldsymbol{J}_{\boldsymbol{\Psi}}^{-1}\dot{\boldsymbol{X}}_R \tag{13}$$

2.2. Governing equations of the numerical model expressed in the referential frame

The proposed model for simulating dynamic crack propagation mechanisms that may lead to macroscopic crack branching inside brittle materials comprises three sets of governing equations: (*i*) the fundamental equations of solid mechanics, (*ii*) the equations governing the kinematics of the moving mesh problem, and (*iii*) the equations related to fracture mechanics, which define conditions for crack onset, propagation development, and crack branching.

Without loss of generality, these equations are introduced herein for a two-dimensional body Ω bounded by an external boundary $\partial\Omega$, which is assumed to be Lipschitz-continuous (Fig. 2-a). In particular, it is assumed that $\partial\Omega$ comprises two portions, named $\partial\Omega_u$ and $\partial\Omega_t$, on which displacements $\boldsymbol{u}(t)=\overline{\boldsymbol{u}}$ and external tractions $\boldsymbol{\sigma}(t)\boldsymbol{n}=\boldsymbol{p}(t)$ are imposed, respectively, such that $\partial\Omega=\partial\Omega_u+\partial\Omega_t$ and $\partial\Omega_u\cap\partial\Omega_t=0$. Initially, the body has an internal crack $\partial\Omega_c$ that develops dynamically with a variable propagation velocity $\dot{a}(t)$. To analyze this behavior, a local coordinate system (x_1^c,x_2^c) is established, with its origin located at the tip O_c of $\partial\Omega_c$. In particular, the local axis x_1^c is aligned tangentially to the crack faces and the propagation direction is identified by the versor $\boldsymbol{n}_c$, which is inclined at an angle $\theta_c(t)$ relative to x_1^c. At a certain time t_b, dynamic instabilities arise inside the body, leading to crack branching. As a result, the initial propagating crack splits into two branches (Fig. 2-b). For each branch, local coordinate systems denoted as $(x_1^{c,1},x_2^{c,1})$ and $(x_1^{c,2},x_2^{c,2})$ are defined, with their origins $(O_{c,1},O_{c,2})$ at the tips of the respective branches.

The branches grow independently, propagating with different propagation velocities and along different directions, namely $\mathbf{n}_{c,1}$ and $\mathbf{n}_{c,2}$, which are identified by kinking angles $\theta_{c,1}$ and $\theta_{c,2}$, respectively. Although the configuration of the body evolves significantly,

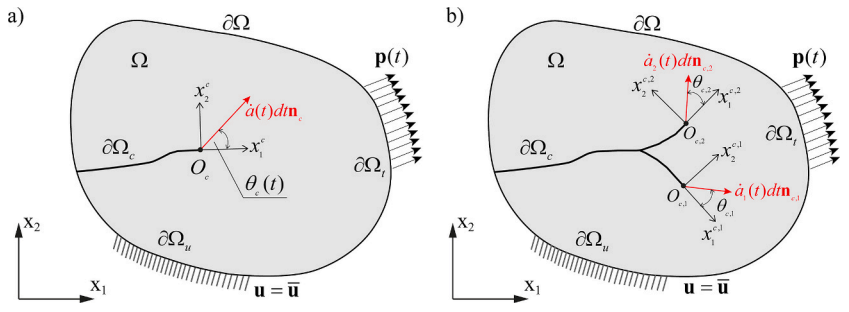


Fig. 2. A schematic of a two-dimensional pre-cracked body under external actions that generate internal dynamic propagation crack mechanisms. Time instances (a) before and (b) after crack branching.

from a mechanical point of view, the governing equations to describe the response of the body before and after the crack branching event are the same.

The weak form of the dynamic equilibrium equation of the body, under the assumption of small displacements and absence of interaction forces along the crack faces, assumes the following form:

$$h\int_{\Omega}[\boldsymbol{C}:\nabla\boldsymbol{u}]:\nabla\delta\boldsymbol{u}\,d\Omega-h\int_{\Omega}\boldsymbol{f}\cdot\delta\boldsymbol{u}\,d\Omega-h\int_{\partial\Omega_p}\boldsymbol{p}(t)\cdot\delta\boldsymbol{u}\,dS=-h\int_{\Omega}\rho\ddot{\boldsymbol{u}}\cdot\delta\boldsymbol{u}\,d\Omega \tag{14}$$

In Eq. (14), **C** represents the fourth-order elastic constitutive tensor, h is the thickness of the body, ρ the mass density, while **f** and $\mathbf{p}(t)$ are the body force and surface traction vectors, respectively. Finally, $\boldsymbol{u}$ and $\delta\boldsymbol{u}$ are the displacement field and a suitable set of virtual displacements, which are defined as follows:

$$\boldsymbol{u}=\left\{\boldsymbol{u}\in H^1(\Omega)\,\middle|\,\boldsymbol{u}=\overline{\boldsymbol{u}}\text{ on }\partial\Omega_u\right\}\qquad\delta\boldsymbol{u}=\left\{\delta\boldsymbol{u}\in H^1(\Omega)\,\middle|\,\delta\boldsymbol{u}=0\text{ on }\partial\Omega_u\right\} \tag{15}$$

where H^1 indicates the Sobolev space and $\overline{\boldsymbol{u}}$ the vector of prescribed displacements on $\partial\Omega_u$.

Note that Eq. (14) is formulated with respect to the material frame (R_X). To obtain the equivalent formulation for the referential frame (R_χ) (*i.e.*, the ALE formulation of Eq. (14)), it is sufficient to express the gradient and time derivatives of the displacement field using Eq. (6) and Eq. (13), thus obtaining:

$$\begin{aligned}&h\int_{\Omega_R}\left\{\left[\boldsymbol{C}:\nabla_R\boldsymbol{u}\boldsymbol{J}_{\boldsymbol{\Psi}}^{-1}\right]:\left(\nabla_R\delta\boldsymbol{u}\boldsymbol{J}_{\boldsymbol{\Psi}}^{-1}\right)\right\}\overline{J}_{\Omega}d\Omega_R-h\int_{\Omega_R}\{\boldsymbol{f}\cdot\delta\boldsymbol{u}\}\overline{J}_{\Omega}d\Omega_R+h\int_{\partial\Omega_R}\{\boldsymbol{p}(t)\cdot\delta\boldsymbol{u}\}\overline{J}_{S}dS_R=\\&-h\int_{\Omega_R}\left\{\rho\left[\ddot{\boldsymbol{u}}_R-2\nabla_R\dot{\boldsymbol{u}}_R\boldsymbol{J}_{\boldsymbol{\Psi}}^{-1}\dot{\boldsymbol{X}}_R-\nabla_R\boldsymbol{u}\boldsymbol{J}_{\boldsymbol{\Psi}}^{-1}\ddot{\boldsymbol{X}}_R+\nabla_R\left[\nabla_R\boldsymbol{u}\boldsymbol{J}_{\boldsymbol{\Psi}}^{-1}\right]\boldsymbol{J}_{\boldsymbol{\Psi}}^{-1}\dot{\boldsymbol{X}}_R\dot{\boldsymbol{X}}_R+\nabla_R\boldsymbol{u}\boldsymbol{J}_{\boldsymbol{\Psi}}^{-1}\nabla_R\dot{\boldsymbol{X}}_R\boldsymbol{J}_{\boldsymbol{\Psi}}^{-1}\dot{\boldsymbol{X}}_R\right]\cdot\delta\boldsymbol{u}\right\}\overline{J}_{\Omega}d\Omega_R\end{aligned} \tag{16}$$

In Eq. (16), $d\Omega_R$ and dS_R are the area and the boundary of the body under the action of external tractions referred to the referential frame. Additionally, $\overline{J}_{\Omega}$ and $\overline{J}_S$ are the Jacobians associated with the volume and area, respectively.

The equations governing the moving mesh problem dictate the motion of the nodes within the computational mesh. Consistent with the ALE formulation, in the present study, these equations follow a rezoning approach based on the Laplace operator applied to the spatial coordinates of mesh nodes in the current configuration, namely $\boldsymbol{x}^T=[x_1\quad x_2]$, *i.e.*:

$$\nabla_R\cdot\nabla_R\boldsymbol{x}=0 \tag{17}$$

The Laplace problem expressed through Eq. (17) must be solved consistently with twofold boundary conditions. The first condition requires that the computational mesh nodes along the external boundary of the body remain fixed (*i.e.*, no displacements). The second condition requires that the mesh node corresponding to the crack tip must advance along the direction of propagation at a velocity equal to the crack propagation velocity $\dot{a}(t)$. By introducing the nodal mesh displacement field $\widetilde{\boldsymbol{u}}=\boldsymbol{x}-\boldsymbol{\chi}$, these boundary conditions can be expressed as follows:

$$\widetilde{\boldsymbol{u}}=\boldsymbol{0}\ \text{ on }\ \partial\Omega;\qquad\dot{\widetilde{\boldsymbol{u}}}\cdot\boldsymbol{n}_c=\dot{a}(t)\ \text{ at }\ O_c \tag{18}$$

Note that the boundary conditions in Eq. (18) define the problem with a single propagating crack. After a crack branching event, additional conditions are needed to describe the growth of new crack tips. Referring to the case depicted in Fig. 2-b, the boundary conditions of the moving mesh problem can be stated as follows:

$$\begin{aligned}&\widetilde{\boldsymbol{u}}=\boldsymbol{0}\ \text{ on }\ \partial\Omega\\&\dot{\widetilde{\boldsymbol{u}}}\cdot\boldsymbol{n}_{c,1}=\dot{a}_1(t)\ \text{ at }\ O_{c,1}\\&\dot{\widetilde{\boldsymbol{u}}}\cdot\boldsymbol{n}_{c,2}=\dot{a}_2(t)\ \text{ at }\ O_{c,2}\end{aligned} \tag{19}$$

More generally, for n_{TIP} crack tips in the computational domain, Eq. (18) and Eq. (19) generalize to:

$$\begin{aligned}&\widetilde{\boldsymbol{u}}=\boldsymbol{0}\ \text{ on }\ \partial\Omega\\&\dot{\widetilde{\boldsymbol{u}}}\cdot\boldsymbol{n}_{c,i}=\dot{a}_i(t)\ \text{ at }\ O_{c,i}\quad\text{For }i=1,..,n_{TIP}\end{aligned} \tag{20}$$

Notably, the propagation direction $\boldsymbol{n}_{c,i}$ and the crack propagation velocity $\dot{a}_i(t)$ can be determined using fracture criteria defined by Fracture Mechanics. These aspects will be clarified later.

The weak form of Eq. (17), including the boundary conditions expressed through Eq. (20), can be expressed using the Lagrangian Multiplier Method (LMM) as follows:

$$\begin{aligned}&\int_{\Omega_R}\left[\nabla_R(\boldsymbol{x})\boldsymbol{J}_{\boldsymbol{\Psi}}^{-1}\right]\cdot\left[\nabla_R(\delta\boldsymbol{x})\boldsymbol{J}_{\boldsymbol{\Psi}}^{-1}\right]\overline{J}_{\Omega}d\Omega_R+\int_{\partial\Omega_R}[\delta(\lambda)\cdot\widetilde{\boldsymbol{u}}+\lambda\\&\cdot\delta(\widetilde{\boldsymbol{u}})]\overline{J}_{\Gamma}dS_R+\sum_{i=1}^{n_{TIP}}\left[\delta(\lambda\cdot\boldsymbol{n}_{c,i})\left(\dot{\widetilde{\boldsymbol{u}}}\cdot\boldsymbol{n}_{c,i}-\dot{a}_i(t)\right)+(\lambda\cdot\boldsymbol{n}_{c,i})\delta\left(\dot{\widetilde{\boldsymbol{u}}}\cdot\boldsymbol{n}_{c,i}\right)\right]_{X=X(O_{c,i})}\\&=0\end{aligned} \tag{21}$$

where, $\boldsymbol{\lambda}$ is the LMM vector and $d\Omega_R$ and dS_R are the infinitesimal volume and surface elements in the referential frame, respectively.

Finally, fracture mechanics equations include criteria to assess crack onset conditions and, more importantly, to evaluate the propagation direction $\theta_{c,i}$ (needed for defining $\mathbf{n}_{c,i}$) and the crack propagation velocity $\dot{a}_i(t)$, which serve in Eq. (20)-b. In the proposed scheme, the crack propagation process is governed using different fracture functions to identify crack onset conditions and to manage the various events involved in the dynamic crack propagation process, such as crack growth, crack arrest, and crack branching.

The fracture function utilized to identify crack onset conditions (f_F^0) is defined consistently with the maximum hoop stress criterion initially proposed by Erdogan and Sin [64]. Precisely, it is defined as follows:

$$f_F^0 = \frac{K^*}{K_{Id}} - 1 \quad \Rightarrow \begin{cases} f_F^0 < 0 & (\text{no } \textit{initiation}) \\ f_F^0 = 0 & (\text{initiation}) \end{cases} \tag{22}$$

where, K_{Id} is the dynamic crack initiation toughness for the material (*i.e.*, the limit threshold for crack onsetting) and K^* represents the equivalent Stress Intensity Factor that, according to Erdogan and Sin [64], can be expressed as follows:

$$K^*(K_I, K_{II}, \theta^*) = K_I \cos^3\left(\frac{\theta^*}{2}\right) - 3K_{II}\cos^2\left(\frac{\theta^*}{2}\right)\sin\left(\frac{\theta^*}{2}\right) \tag{23}$$

In Eq. (23), (K_I, K_{II}) are the mode I and mode II Dynamic Stress Intensity Factors (DSIFs) at the crack front and θ^* is the kinking angle evaluated with respect to the crack tip system of coordinates (x_1^c, x_2^c). Consistent with the maximum hoop stress criterion, the kinking angle is expressed as follows:

$$\theta^*(K_I, K_{II}) = 2\arctan\left[\frac{K_I}{4K_{II}} - \frac{sign(K_{II})}{4}\sqrt{\left(\frac{K_I}{K_{II}}\right)^2 + 8}\right] \tag{24}$$

From Eqs. (23)–(24), it is evident that the evaluation of crack onset conditions and the crack propagation angle requires an accurate evaluation of DSIFs at the crack front. In the proposed scheme, the DSIFs are evaluated using the $M-$integral method, which represents a practical and reliable approach to computing such variables in the context of computational fracture mechanics. In particular, the proposed approach uses the ALE formulation of the $M-$integral. For the sake of brevity, the ALE formulation of the $M-$integral is not reported in the present work. However, a comprehensive discussion on the subject is provided in [59].

The fracture function governing the crack growth process (f_F^p) serves to obtain the crack propagation velocity and assess crack arrest conditions. It is defined as follows:

$$f_F^p = \frac{K^*}{K_{ID}} - 1 \Rightarrow \begin{cases} f_F^p = 0 \ \ \& \ \ \dot{a} > 0 & (\text{propagation}) \\ f_F^p = 0 \ \ \& \ \ \dot{a} = 0 & (\text{arrest}) \end{cases} \tag{25}$$

where $K_{ID} = K_{ID}(\dot{a})$ is the dynamic crack growth toughness of the material, which is a parameter dependent on the crack propagation velocity. Following experimental observations, Kanninen and Popelar [65] proposed the following empirical expression to define $K_{ID}(\dot{a})$:

$$K_{ID}(\dot{a}) = \frac{K_{IA}}{1 - \left(\frac{\dot{a}}{V_L}\right)^{\overline{m}}} \tag{26}$$

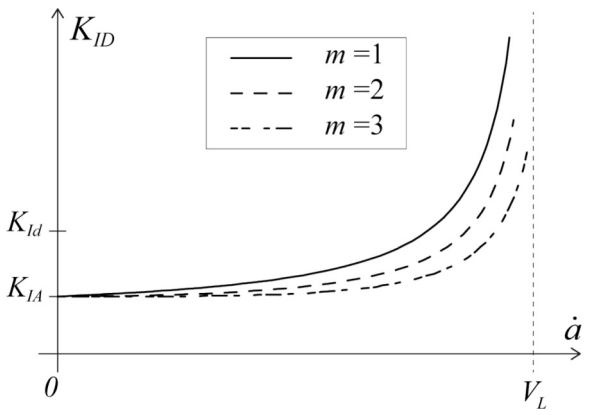


Fig. 3. A representation of the equation initially proposed by Kanninen and Popelar to schematize the variability of the dynamic crack growth toughness as a function of the crack propagation velocity.

In Eq. (26), K_{IA} represents the crack arrest toughness, which is a fundamental material property that characterizes the ability of the material to arrest a crack propagation event. Additionally, V_L is the limiting crack propagation velocity, and m is a dimensionless shape parameter inherent to Eq. (26) that governs the shape of the dynamic crack growth toughness (K_{ID}) curve. Given its significant influence on the shape of this curve, m can also be considered a material constant. Fig. 3 shows a representation of the Eq. (26) for increasing values of the shape parameter m.

This graph denotes that K_{ID} has slight variation for low crack tip velocities. In particular, it results $K_{ID}(\dot{a} = 0) = K_{IA} = 0$. On the other hand, it increases for larger propagation velocity values according to a nonlinear trend, approaching an infinite value as $\dot{a}$ tends to V_L. In addition, Fig. 3 shows that K_{Id} is higher than K_{IA}, meaning that the energy that has to be spent to onset the crack propagation is generally higher than the one required to grow the crack. Eqs. (25)–(26) are essential in the proposed scheme to evaluate the crack propagation velocity that has to be introduced in Eq. (20)-b.

2.3. Fracture functions for crack branching

The proposed method belongs to the class of numerical strategies that require a suitable criterion to determine the crack branching time [5].

The literature reports three fundamental criteria for establishing the branching time: (*i*) the velocity criterion, (*ii*) the stress intensity factor criterion, and (*iii*) the energy criterion. The proposed method offers substantial flexibility, allowing for a straightforward implementation of any of these criteria to accurately determine the crack branching time. In the following, attention is focused on the first two criteria, namely the velocity criterion and the stress intensity factor criterion, with their corresponding fracture functions explicitly defined.

The velocity criterion uses the crack propagation velocity as the reference parameter, stating that branching occurs when the velocity reaches a critical value (a_{cr}). It is worth noting that experimental findings reveal two types of branches, *i.e.*, micro-branches and macro-branches, which generally occur at different crack propagation velocities. For these reasons, the literature proposes distinct branching criteria for micro- and macro-branches.

Micro-branches develop at micro-scales when the velocity of a propagating crack reaches a critical value that determines unstable crack propagation. In this condition, the propagation velocity is affected by large oscillations, which cause surface roughening and the formation of microbranches. Critical velocities for micro-branching have been identified for various materials, typically expressed as a function of the material's Rayleigh wave speed (c_R). Thus, for instance, Hauch and Marder [66] found that the critical velocity for Homalite is equal to 0.37 c_R, while Gross et al. [67] determined that it is equal to 0.42 c_R for soda-lime glass.

Macro-branches originate from the coalescence of voids and micro-branches inside the material. Experimental observations helped define the critical velocity for macro-branching in glass (0.5–0.65 cR), PMMA (0.6–0.7 cR), and Homalite-100 (0.35–0.45 c_R). Additionally, various theoretical models have been proposed for macrobranching. Among these, the model proposed by Yoffe et al. [3] indicates that when the propagation velocity exceeds the critical threshold of 0.6 c_S (where c_S is the shear wave speed of the material), the propagation process becomes unstable, resulting in branching. In particular, the branching cracks develop along directions inclined at 60° regarding the original crack trajectory.

In the context of computational fracture mechanics, the velocity criterion has achieved notable success due to its simplicity in implementation. Consequently, many numerical models employ this criterion to simulate crack branching phenomena (see, for instance, [68–70]).

Notably, the proposed model only simulates the fracture process associated with the occurrence of macro-branching. Assuming a velocity criterion, the fracture function related to crack branching is defined as follows:

$$f_F^{B,v} = \frac{\dot{a}(t)}{a_{cr}} - 1 \Rightarrow \begin{cases} f_F^{B,v} = 0 & \text{(branching)} \\ f_F^{B,v} < 0 & \text{(stable propagation)} \end{cases} \tag{27}$$

The stress intensity factor-based criterion states that branching occurs when the DSIFs reach a critical value. Based on experimental observations, Kobayashi and Mall [71] and Dally [12] defined a critical fracture toughness for branching whose value could be assumed as two or three times greater than the quasi-static fracture toughness of the material. Rafiee et al. [72] have efficiently implemented this criterion in a BEM-based numerical model to simulate crack curving and branching phenomena inside brittle materials. Kishen and Singh [73] employed the stress intensity factor-based criterion in an FE-based model to study the behavior of concrete dams.

In the context of the proposed method, the stress intensity factor criterion can be implemented by defining the following fracture function:

$$f_F^{B,K} = \frac{K^*}{K_b} - 1 \Rightarrow \begin{cases} f_F^{B,K} = 0 & \text{(branching)} \\ f_F^{B,K} < 0 & \text{(stable propagation)} \end{cases} \tag{28}$$

where K^* is the equivalent stress intensity factor (Eq. (23)) and K_b is the critical stress intensity factor for branching of the material.

3. Numerical implementation

The governing equations of solid mechanics (Eq. (16)) and those of the moving mesh problem (*i.e.*, the rezoning approach described

by Eq. (21)) have been implemented in a user-modified version of COMSOL Multiphysics [74], a commercial FE-based multiphysics software. According to the FE method, the geometry domain is subdivided into a finite number of elements (N^e), and the accuracy of the solution directly depends on the number of elements used. In the context of the proposed approach, the primary variables, *i.e.*, the displacement field (**u**), the position of the nodes of the computational mesh (**x**), and Lagrangian multipliers (λ), are evaluated at the nodes of the computational mesh. The solution fields inside the computational domain are then approximated.

The approximated solutions are expressed as the product of the nodal variables and isoparametric shape functions, as follows:

$$\boldsymbol{u} = \sum_{i=1}^{N^e} \boldsymbol{N}_i^U \boldsymbol{u}_{L,i}; \quad \overline{\boldsymbol{x}} = \sum_{i=1}^{N^e} \boldsymbol{N}_i^{ALE} \overline{\boldsymbol{x}}_{L,i}; \quad \lambda = \sum_{i=1}^{N^e} \boldsymbol{N}_i^{\lambda} \lambda_{L,i} \tag{29}$$

In Eqs. (29), $(\boldsymbol{u}_{L,i}, \boldsymbol{x}_{L,i}, \lambda_{L,i})$ represent the vectors of nodal variables of the displacement fields, nodal position, and Lagrangian multiplier associated with the i-th node, whereas $(\boldsymbol{N}_i^U, \boldsymbol{N}_i^{ALE}, \boldsymbol{N}_i^{\lambda})$ are the matrices containing the interpolation functions of the finite elements. By introducing Eq. (29) in Eq. (16) and Eq. (21), the following system of equations results:

$$\begin{aligned} &\boldsymbol{M}\ddot{\boldsymbol{u}}_G + \boldsymbol{K}\boldsymbol{u}_G = \boldsymbol{f} \\ &\boldsymbol{A}\overline{\boldsymbol{x}}_G + \boldsymbol{r}\lambda_G = \boldsymbol{0} \end{aligned} \tag{30}$$

Here, the subscript G indicates that the nodal variables are global. Additionally, in the first line, **M** and **K** are the mass and stiffness matrix of the assembled domain, respectively, while **f** is the vector containing the external actions. Finally, in the second line, **A** and **r** represent the matrix of coefficients and the vector referred to the MM problem, respectively.

The proposed approach manages the crack propagation process through a customized algorithm designed to handle dynamic crack propagation processes. The algorithm consists of well-defined steps that govern crack evolution and mesh variation at each time step. Fig. 4 shows a flowchart summarizing the sequence followed by the developed algorithm to simulate dynamic fracture propagation inside a generic brittle material, whereas Fig. 5 provides schematic illustrations of some key steps of the procedure.

Beyond some preliminary operations concerning essential data input, such as material properties and geometry configuration, the algorithm automatically executes all remaining steps. It is worth noting that the proposed approach can simulate the fracture behavior of a pre-cracked body only, which requires that the geometric configuration of the domain under investigation must include a geometric entity representing an internal discontinuity (Fig. 5-a).

After entering the input data, preliminary operations are performed on the geometric entities representing the internal pre-crack.

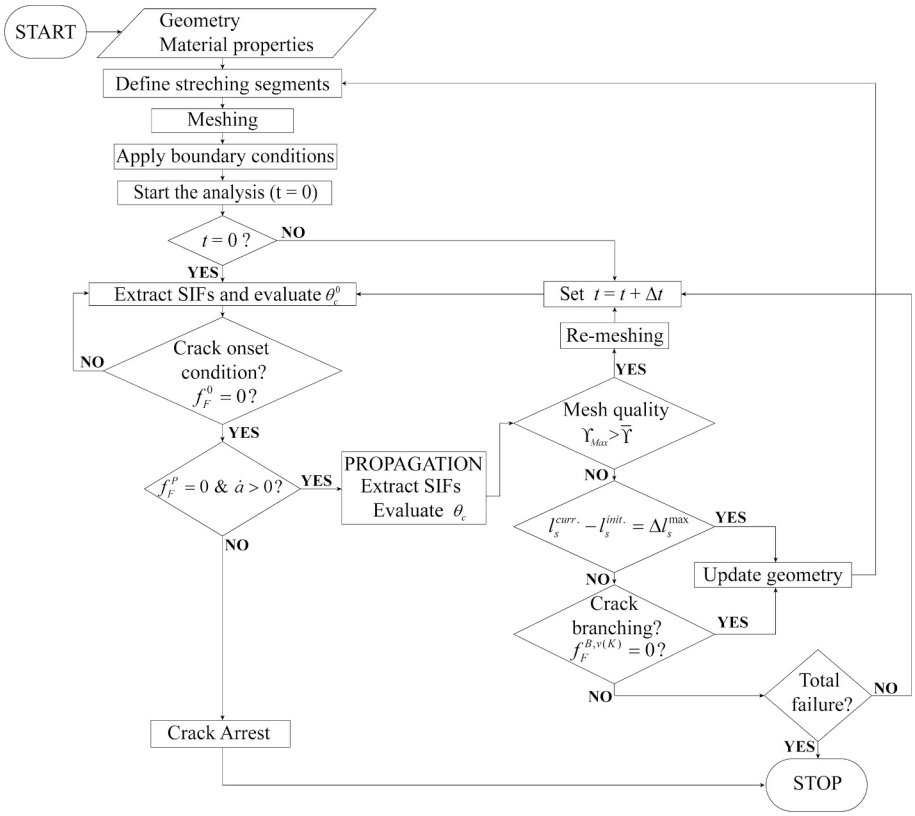


Fig. 4. Flowchart depicting the steps that the proposed model follows to simulate crack propagation processes.

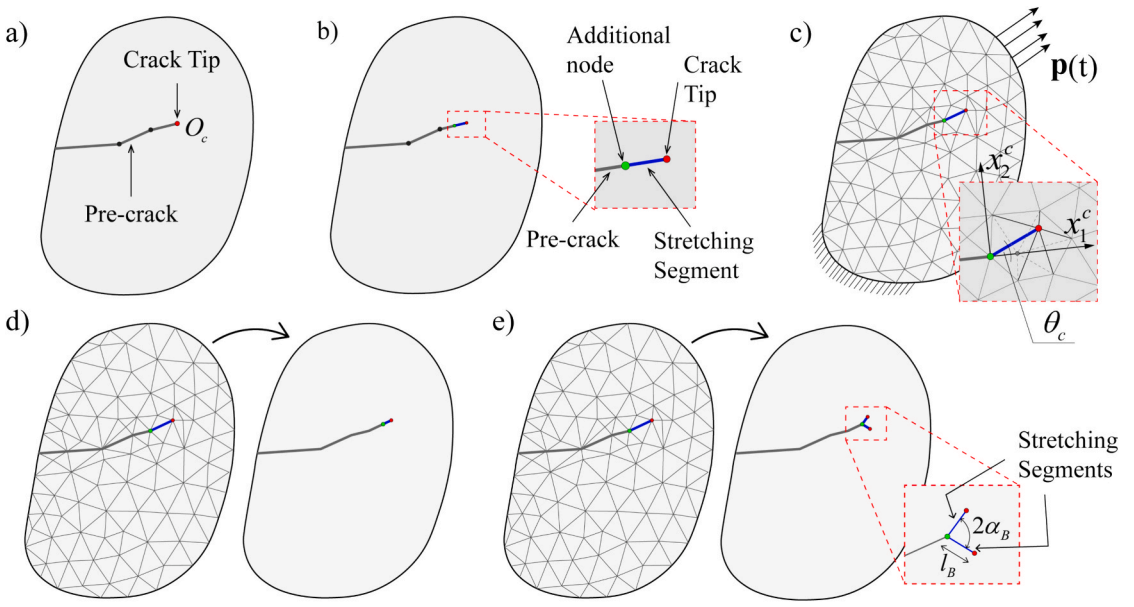


Fig. 5. A schematic of some of the steps followed by the proposed model for reproducing the crack propagation process: a) input geometry; b) creation of the stretching segment; c) crack propagation phase with a zoomed view; d) geometry update when the maximum length of the stretching segment is reached; e) geometry update in case of crack branching.

These operations involve the insertion of an additional node near the crack tip, thus dividing the pre-crack into two portions that have different scopes (Fig. 5-b). The longest portion, which extends from the external boundary to the added node, remains fixed during propagation and acts as the effective initial pre-crack. On the other hand, the shorter portion, referred to as the stretching segment, is subjected to elongation and rotation using the moving mesh technique, thereby effectively reproducing dynamic crack growth during the analysis. The length of the stretching segment is typically determined based on material properties and crack growth conditions, ensuring a smooth progression of the crack.

Once the stretching segment is defined, the computational domain is meshed. In particular, the mesh is refined around the stretching segment region to achieve adequate accuracy in the $M-$integral calculation (essential for reliable DSIFs evaluation). In contrast, a coarser mesh is adopted elsewhere in the domain.

Following the mesh definition, boundary conditions are imposed, and the analysis starts. At each time step of the simulation, the DSIFs are evaluated using the $M-$integral method together with an initial kinking angle θ_c^0 (calculated through Eq. (24)). Concurrently, the initiation fracture function f_F^0 (Eq. (22)) is checked, and if $f_F^0 = 0$, the propagation phase begins.

During propagation, the process is controlled by the propagation function f_F^P. In particular, the computational node associated with the crack tip advances along the direction of propagation at a crack propagation velocity that satisfies the condition $f_F^P = 0$ (Eqs. (25)–(26)) (see Fig. 5-c). It is essential to note that the propagation direction is recalculated at each time step of the simulation, thereby allowing for dynamic mesh adjustments in response to evolving fracture conditions.

The movement of the mesh node associated with the crack tip inevitably causes distortions in the surrounding finite elements because the MM approach relies on adaptive node repositioning rather than continuous remeshing actions. These distortions can negatively affect the reliability and computational stability of the proposed strategy. For this reason, the quality of the finite elements inside the computational mesh is continuously monitored during the propagation phase.

To quantify finite element distortions, the first invariant of the isochoric Green-Lagrange strain (here denoted by Υ) is calculated for all the finite elements in the computational mesh. By setting a threshold value for Υ (namely, $\overline{\Upsilon}$), the code evaluates the maximum value (Υ_{Max}) at each time step of the propagation phase. If $\Upsilon_{Max} \geq \overline{\Upsilon}$ a remeshing of the computational mesh is performed. This condition denotes that, unlike the standard FE-based procedures that execute a remeshing of the computational domain at every incremental crack growth, the proposed approach executes a remeshing action exclusively when finite element distortions become excessive, thereby reducing computational costs considerably.

Besides the quality of finite elements, the extension of the stretching segment and the occurrence of crack branching are also monitored. The first check aims to limit the current length of the stretching segment ($l_s^{curr.}$) relative to its initial value ($l_s^{init.}$). Specifically, by setting a maximum extension for the stretching segment (namely $\Delta l_s^{\max}$), the code verifies whether $l_s^{curr.}$ reaches the predefined threshold (*i.e.*, $l_s^{curr.} - l_s^{init.} = \Delta l_s^{\max}$). If this condition is met, the analysis is paused, and a new geometry is created based on the mesh configuration at the time of pause. A new stretching segment is then created in the novel geometry. After meshing it, the analysis resumes, adopting the displacement solutions from the step immediately preceding the pause as initial conditions for the subsequent simulation (Fig. 5-d). It is worth noting that assessing the stretching segment length is essential to ensure realistic curved crack paths

during propagation, as a series of linear segments approximates the curve. As a consequence, small values of the threshold value $\Delta l_s^{\max}$ ensure smooth crack trajectories.

The occurrence of crack branching is assessed using the branching fracture function (defined by Eq. (27) or Eq. (28)). Similar to the condition concerning the stretching segment length, if $f_F^{B,v} = 0$ (or equivalently $f_F^{B,K} = 0$), the analysis is paused and a new geometry for the computational domain is created based on the mesh configuration at the moment of pause. In the novel geometry, two short segments representing crack branches are introduced at the crack front (*i.e.*, the splitting of the primary crack front). In particular, the branching segments have a length l_b and are inclined at an angle α_B with respect to the original crack plane (Fig. 5-e). The new geometry configuration is meshed, and the analysis restarts. In line with the proposed strategy, the branching segments work as stretching segments once the analysis resumes. It is worth noting that both such parameters are user-defined. In this context, l_b must be set sufficiently small to ensure reliable results while maintaining computational stability. For α_B, a preliminary trial value between 10° and 45° (in line with experimental evidence [5]) can be selected. However, a more precise value for α_B may be defined following theoretical criteria reported in the literature [3,75–78]. In this context, Ramulu and Kobayashi [79] report that the crack branching direction α_B can be identified using the maximum circumferential stress [64] or the minimum strain energy density [80] criterion.

In the proposed modeling strategy, the propagation direction of the branching segments is determined according to the maximum circumferential stress criterion. Because the branching segments act as stretching segments in the steps immediately after crack branching, the moving mesh technique dynamically adjusts their orientation consistently with the computed crack propagation directions, thus automatically correcting the trial value of α_B. As a consequence, by leveraging the MM technique's capabilities, the need for explicitly defining specific criteria for determining the crack branching angle is avoided, thereby simplifying numerical implementation. This represents yet another advantage of using the moving mesh technique for modeling crack propagation processes in brittle materials compared to traditional FE-based strategies.

The numerical analysis continues following the steps reported in Fig. 4 until the total collapse of the body under investigation occurs.

4. Results

Three relevant benchmark cases are analyzed to assess the reliability and efficiency of the proposed approach in reproducing dynamic crack propagation processes leading to crack branching phenomena. These cases were chosen for their significance in computational fracture mechanics, providing a robust framework for validating numerical strategies against experimental and numerical results.

The first case consists of a rectangular plate with a horizontal initial notch under tension, which has been extensively studied using various numerical methods. Since literature data regarding the crack path and the time history of crack tip velocity are available, this case offers the opportunity to assess the accuracy of the proposed method from multiple perspectives.

The second case involves a square plate with a horizontal initial notch subjected to biaxial loading. This configuration introduces additional complexity due to the inherent influence of biaxial stress conditions on crack initiation and propagation. The availability of both experimental and numerical results in the literature makes this a highly relevant benchmark for validating the proposed strategy.

Finally, the third case investigates the fracture behavior of a plate under mixed-mode dynamic crack propagation with branching. Therefore, this case represents a suitable benchmark for assessing the capabilities of the proposed model in predicting crack branching phenomena under mixed-mode conditions.

The results obtained from the proposed strategy are compared with data available in the literature for all study cases. In particular, numerical simulations conducted using this method employed computational meshes of triangular finite elements with quadratic interpolation functions.

For temporal discretization, numerical simulations were conducted using an adaptive time-stepping scheme. An implicit Backward Differentiation Formula (BDF) solver was specifically employed, which adjusts the time step dynamically throughout the simulation. This adaptive approach ensures that the specified accuracy tolerances, both relative and absolute, are preserved during the entire simulation process. To address the potential damping effects commonly associated with lower-order BDF methods, the adaptive solver primarily utilized higher-order BDF formulations (up to order five) during critical phases of the simulation. This choice accurately

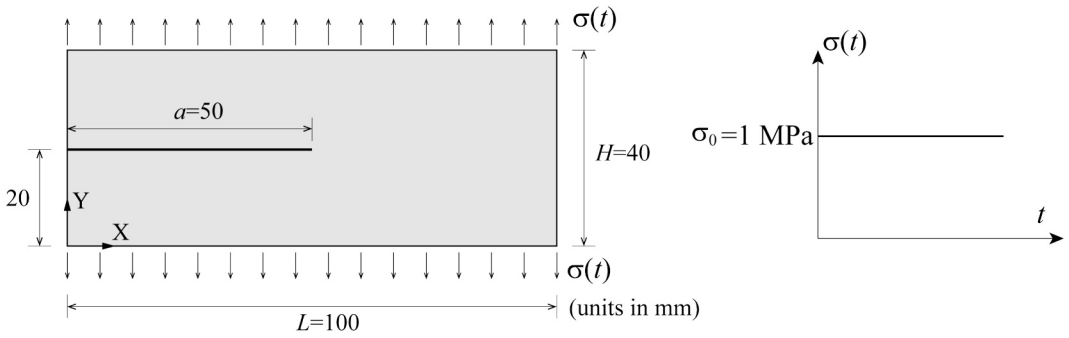


Fig. 6. A rectangular plate with a horizontal initial notch under tension. Geometry and loading conditions.

represents sharp gradients and high-frequency phenomena, particularly those related to dynamic crack propagation. The BDF method is particularly effective for dynamic fracture problems where rapid changes in the solution—such as crack initiation and propagation—require a flexible temporal resolution. This adaptability guarantees the robustness and reliability of the results obtained. A maximum time step was also established for data output to facilitate reporting and visualization, ensuring a sufficiently dense dataset for post-processing.

4.1. A rectangular plate with a horizontal initial notch under tension

Fig. 6 shows a rectangular plate of length $L = 100$ mm, height $H = 40$ mm, and thickness $B = 1$ mm subjected to uniform traction $\sigma(t) = \sigma_0 = 1$ MPa acting along the upper and bottom boundaries. Besides, it has a horizontal pre-crack $a = 50$ mm originating from the left boundary and extending to the center.

The material is glass, with Young's modulus, Poisson ratio, and mass density equal to $E = 32$ GPa, $\nu = 0.2$, and $\rho = 2450$ kg/m^3, respectively. According to these material properties, the dilatational wave speed, the shear wave speed, and the Rayleigh wave speed are equal to $c_D = 3809.5$ m/s, $c_S = 2332.8$ m/s, and $c_R = 2119$ m/s, respectively. Additionally, the dynamic crack initiation toughness is given by $K_{Id} = 0.33$ MPa m$^{1/2}$.

Fig. 7 illustrates the computational mesh configuration used by the proposed method at the beginning of the simulation. The mesh consists of 584 triangular elements, which are densely arranged in the region around the crack tip and coarsely in the remaining part of the computational domain. This discretization arrangement allows for an accurate evaluation of fracture parameters near the singularity zone. It is also worth noting that the proposed modeling strategy has already been subjected to a detailed mesh sensitivity analysis in a previous study by some of the present authors [59]. The results demonstrated that the proposed method provides robust and accurate evaluation across different mesh resolutions, particularly concerning the evaluation of DSIFs during mesh motion.

The zoomed view of the crack tip region illustrated in Fig. 7 shows a detail of the stretching segment, which has a length of 2.5 mm and is discretized using 3-line elements. The numerical analysis is performed under plane stress conditions, which are appropriate for thin plates (thickness = 1 mm).

This case has been investigated numerically by various authors through different numerical strategies, including FEM [81], X-FEM [33,82,83], phase field [46,55,84], and meshless [40,85] methods, together with cohesive zone models [86], and peridynamics model [49,52].

The velocity criterion was utilized to determine the crack branching time to reproduce the fracture behavior of the plate using the proposed method. In particular, a critical velocity of $a_{cr} = 0.6c_R$ was assumed, which is consistent with experimental observations and numerical methods based on internal crack branching criteria [5]. Besides, for reporting and visualization purposes, a maximum time step for data output was set at 0.1 μs, ensuring a sufficiently dense dataset for post-processing.

4.1.1. Calibration of the dynamic crack growth toughness curve of the material

A crucial aspect of the proposed methodology is the selection of appropriate values for the crack arrest toughness (K_{IA}), the dimensionless shape parameter (m), and the limiting crack propagation speed (V_L), as these directly determine the dynamic crack growth toughness (K_{ID}) of the material according to Eq. (26).

The K_{ID} curve is a key parameter that significantly affects the fracture behavior of the material under dynamic conditions, including critical aspects like crack propagation velocity.

Typically, V_L is assumed to be equal to the Rayleigh wave speed of the material, representing the theoretical upper limit for crack propagation velocity in brittle materials. Therefore, for this study, V_L is set at $c_R = 2119$ m/s. However, precise values of K_{IA} and m are not readily available in the literature for the specific material in question. Consequently, these parameters were determined through a systematic calibration analysis based on the consistent numerical observation indicating that crack branching initiates when the crack extends approximately 200 mm. This result, observed across various numerical methods, served as a pivotal benchmark for the calibration, allowing for the inference of suitable values for K_{IA} and m.

For brittle materials, K_{IA} is generally recognized to be lower than the crack initiation toughness (K_{Id}). Therefore, a broad range of admissible values for K_{IA} was considered in the calibration analysis, specifically between 0.5 K_{Id} and 0.9 K_{Id}. For the dimensionless shape parameter m, a range between 2 and 5 was assumed, which encompasses typical values for this type of material behavior found

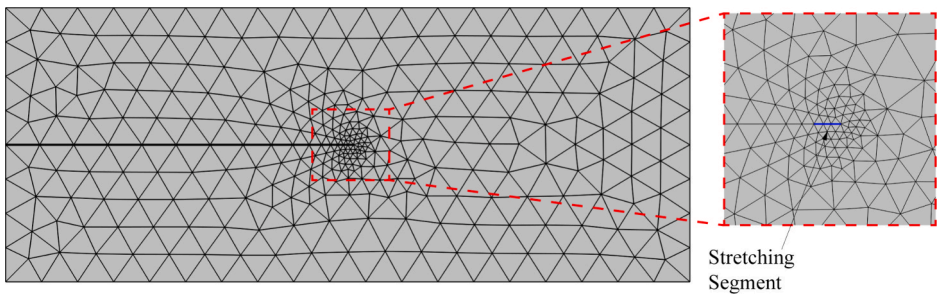


Fig. 7. A rectangular plate with a horizontal initial notch under tension. Initial mesh configuration.

in the literature.

The calibration procedure was conducted iteratively by progressively narrowing the parameter ranges to identify the "optimal" values of K_{IA} and m.

Initially, numerical simulations were performed using combinations of extreme values for K_{IA} and m within their respective ranges. Specifically, the cases considered were: (*i*) $K_{IA} = 0.9\ K_{Id}$ and $m = 2$ (curve c_1 in Fig. 8-a, describing a condition of maximum fracture dynamic toughness), and (*ii*) $K_{IA} = 0.5\ K_{Id}$ and $m = 5$ (curve c_2 in Fig. 8-a, representing a condition of minimum dynamic fracture toughness). Fig. 8-b shows the evolution of the dimensionless crack tip velocity ($\dot{a}/0.6c_R$) as a function of the dimensionless crack tip position (x_{Tip}/L) corresponding to curves c_1 and c_2. It is evident that neither of these initial combinations accurately reproduced the target results. In particular, these led to a fracture behavior of the plate that deviated significantly from expectations.

Next, an intermediate combination of values was considered by assuming $K_{IA} = 0.7\ K_{Id}$ and $m = 3$ (curve c_3, see Fig. 8-a). Fig. 8-b shows that the associated K_{ID} curve results in a fracture behavior of the material plate for which the maximum propagation velocity approaches $\dot{a}/0.6c_R = 0.97$ at $x_{Tip}/L = 0.73$, which is close to the target result. Fine-tuning the previous values defines a fourth curve with $K_{IA} = 0.7\ K_{Id}$ and $m = 5$ (curve c_4 in Fig. 8-a), which lies between curves c_2 and c_3. This combination allows for a suitable reproduction of the expected target for crack branching. These values are assumed in the subsequent numerical simulations devoted to validating the proposed methodology.

It is important to highlight that while these parameters matched the branching condition well, other combinations of (K_{IA}, m) that yield K_{ID} curves sufficiently close to curve c_4 would also be acceptable for modeling the material's dynamic fracture behavior. Indeed, curve c_4 is likely to be very close to the effective dynamic fracture toughness curve of the material.

To further substantiate this claim and demonstrate the robustness of the proposed method concerning parameter sensitivity, additional analyses were conducted with alternative K_{ID} curves closely resembling curve c_4. Specifically, three new combinations of parameters were investigated: (*i*) $K_{IA} = 0.7\ K_{Id}$, $m = 4$ (curve c_5 in Fig. 9-a), (*ii*) $K_{IA} = 0.8\ K_{Id}$, $m = 5$ (curve c_6 in Fig. 9-a), and (*iii*) $K_{IA} = 0.6\ K_{Id}$, $m = 4$ (curve c_7 in Fig. 9-a). The resulting curves are very close to curve c_4 and were used to simulate the fracture behavior of the plate.

Fig. 9-b presents graphs of the dimensionless crack tip position versus dimensionless propagation velocity. The results are highly comparable, demonstrating that the overall fracture behavior, including crack branching, remains consistent even with minor variations in the calibrated material parameters.

These results suggest that the K_{IA} and m values determined through the calibration procedure are very close to the effective material properties. More precisely, it is likely that the effective material properties fall within the ranges of $0.6 \leq K_{IA} \leq 0.8$ and $4 \leq m \leq 5$, as different combinations of these parameters provide similar dynamic crack growth toughness (K_{ID}) curves. This consistency is further supported by the fundamental nature of these parameters as material constants. Specifically, K_{IA} is a material property governing crack arrest, while m is a dimensionless parameter (inherent to the model) that crucially defines the rate-dependent shape of the dynamic crack growth toughness (K_{ID}) curve, thus intrinsically reflecting a material characteristic.

Additionally, it is crucial to emphasize that K_{IA} and m not only affect the crack propagation behavior during the pre-branching phase but also significantly influence the crack propagation direction and velocity in the post-branching phase. If these values are incorrectly estimated, the resulting crack paths and velocities in the post-branching phase could be unreliable, potentially compromising the validity of the simulation results.

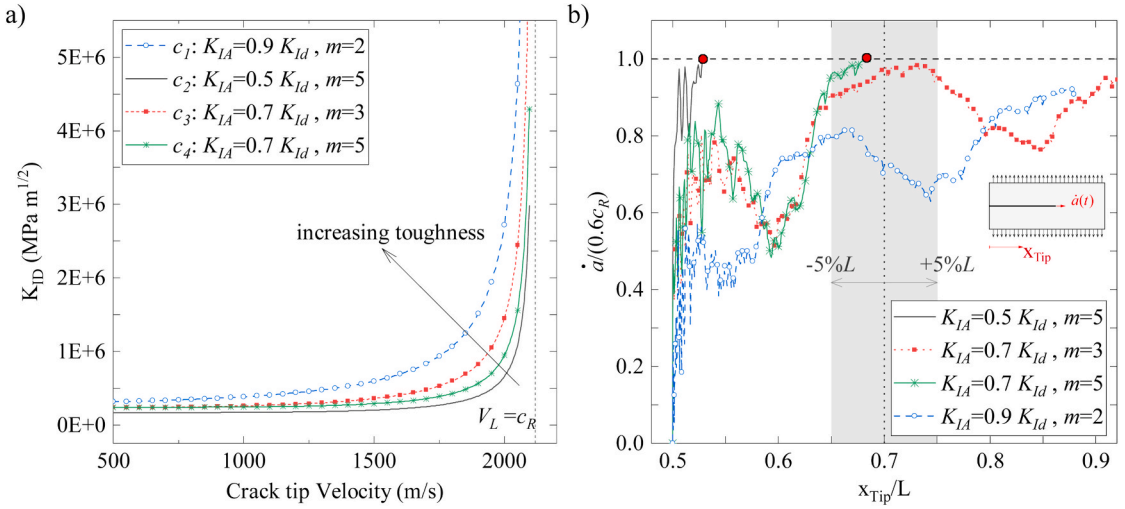


Fig. 8. A rectangular plate with a horizontal initial notch under tension. The calibration procedure for the crack arrest toughness and the dimensionless shape parameter: (a) Dynamic crack growth toughness curves. (b) Evolution of the dimensionless crack tip propagation velocity as a function of the dimensionless crack tip position.

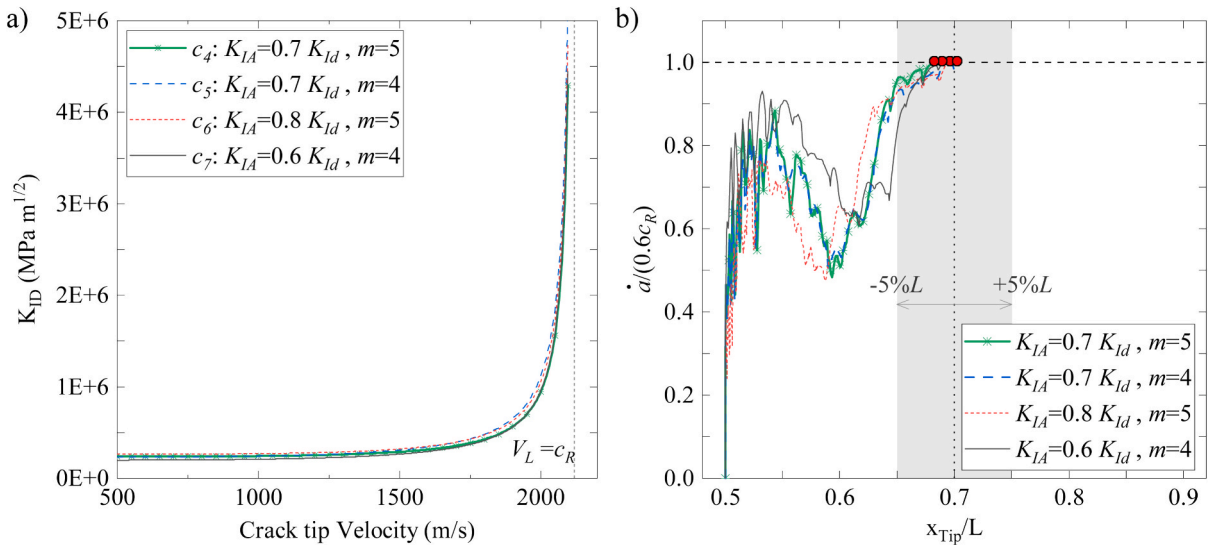


Fig. 9. A rectangular plate with a horizontal initial notch under tension. (a) Dynamic crack growth toughness curves close to curve c_4. (b) Evolution of the dimensionless crack tip propagation velocity as a function of the dimensionless crack tip position.

4.1.2. Validation of the proposed approach

Fig. 10 compares the crack tip propagation velocity predicted by the proposed strategy with results obtained using various numerical methods, which include: (*i*) the Phase-Field regularized Cohesive Zone Model (PF-CZM) proposed by Nguyen and Wu [55]; (*ii*) a peridynamic model developed by Agwai et al. [49]; (*iii*) a rainfed FE-based approach employing the Cohesive Zone Model [86]; and (*iv*) an XFEM-based method enriched with meshfree approximations, proposed by Menouillard and Belytschko [82]. Notably, among these methods, only the latter employs an external criterion to assess the crack branching time. However, unlike the velocity criterion, it adopts the energetic criterion initially proposed by Döll [87].

Fig. 11(a)-(f) provide visual insights into the crack propagation and branching process, illustrating the evolution of the computational mesh at increasing time instants during the numerical simulation.

According to the proposed method, the crack propagation process begins 10.3 μs after the start of the simulation. From this instant, the crack propagates horizontally at a velocity that evolves nonlinearly, as depicted in Fig. 10. Specifically, the velocity initially

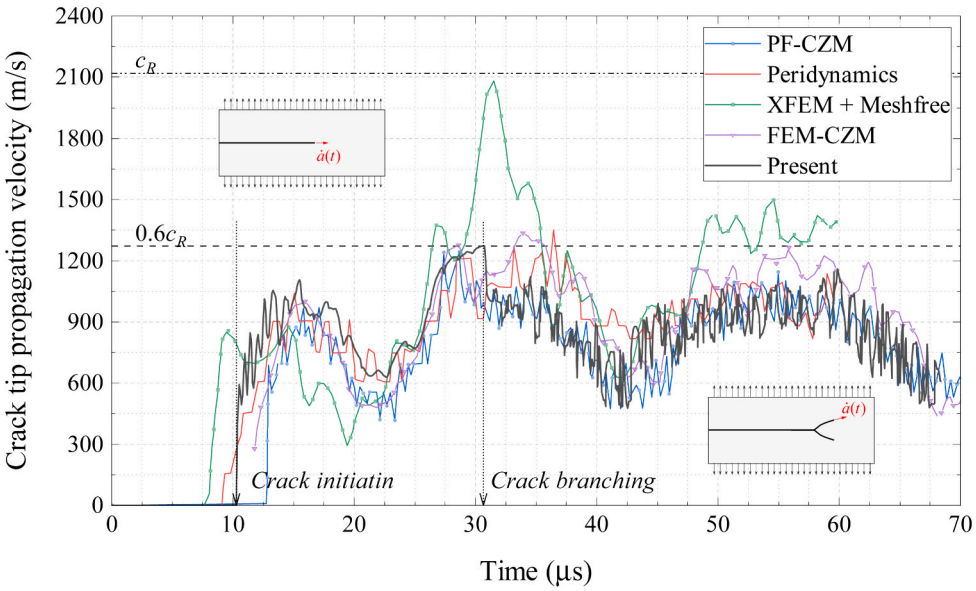


Fig. 10. A rectangular plate with a horizontal initial notch under tension. Time history of a crack tip propagation velocity: comparison between the proposed approach and predictions of Nguyen and Wu [55] (Phase-Field regularized Cohesive Zone Model (PF-CZM)), Agwai et al. [49] (Peridynamics), Menouillard and Belytschko [82] (XFEM + Meshfree), and Hirmand and Papoulia [86] (FEM-CZM).

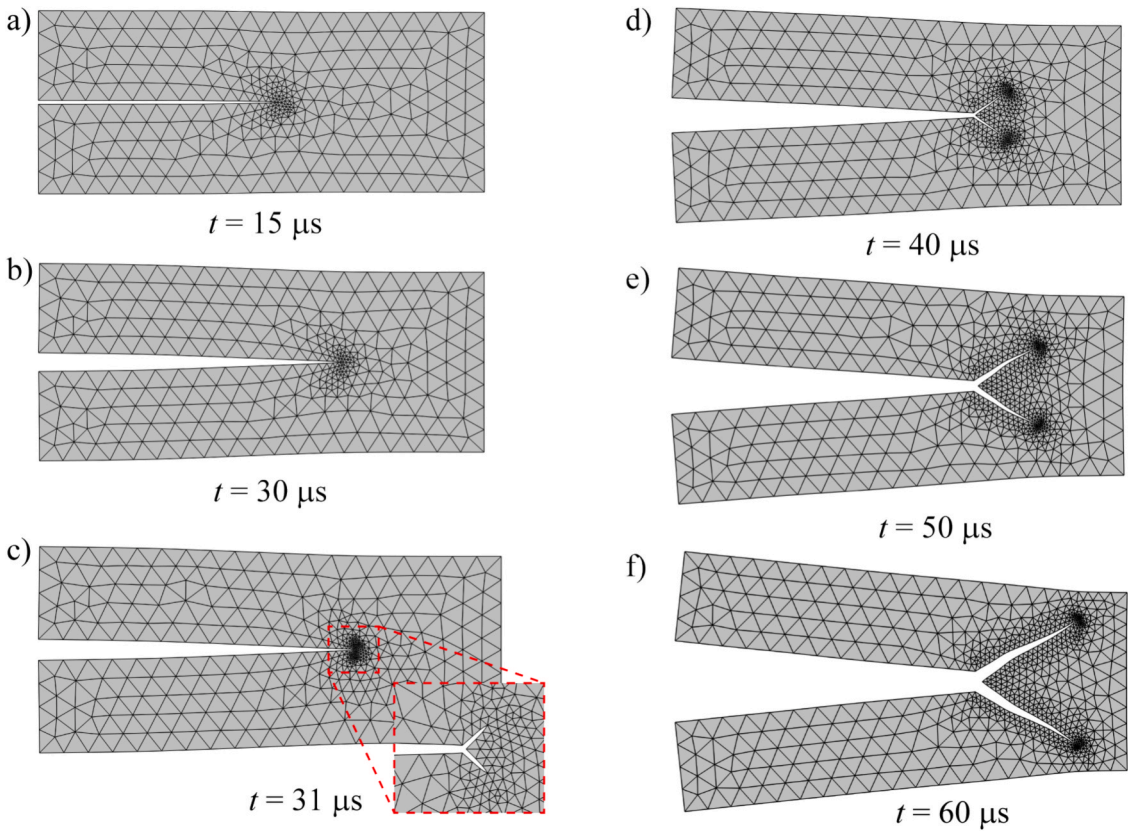


Fig. 11. A rectangular plate with a horizontal initial notch under tension. Evolution of the computational mesh at different instants: (a) t = 15 μs; (b) t = 30 μs; (c) t = 31 μs; (d) t = 40 μs; (e) t = 50 μs; (f) t = 60 μs.

increases sharply to 1080 m/s at $t = 15$ μs, then decreases to approximately 600 m/s over the next 10 μs. Subsequently, it rises again, reaching the threshold value of 0.6 c_R (1271 m/s) at $t = 30.5$ μs, which marks the crack branching time.

At crack branching time ($t_B = 30.5$ μs), the computational domain is updated in accordance with the developed crack propagation strategy. A new geometry is generated based on the mesh configuration at the time of crack branching. This operation involves inserting two additional segments that depart from the crack tip to reproduce the splitting of the crack trajectory. Fig. 11-b and c illustrate snapshots of the computational mesh immediately before and after the crack branching time. For the case under discussion, the additional segments have a length $l_B = 1.25$ mm, which is a very small extension relative to the overall computational domain and the initial crack length. Besides, these segments are symmetrically inclined at $\alpha_B = 45^\circ$ relative to the previous crack trajectory.

Once the crack trajectory splits, the propagation phase resumes. The newly formed cracks propagate along nearly symmetric trajectories with respect to the horizontal. For clarity, Fig. 10 reports only the propagation velocity of the upper crack (*i.e.*, the crack that proceeds toward the upper right corner of the plate after branching), as the propagation velocities of both cracks are quite similar during the simulation. As observed, the evolution of the crack propagation velocity after the crack branching time aligns very well with the results obtained through the PF-CZM and the FEM-CZM methods. These methods predict a decreasing trend up to an average value of 600 m/s around $t = 40$–43 μs, followed by a parabolic-like evolution until the end of the simulation at $t = 160$ μs. The peridynamic model exhibits a similar trend, although with a delayed crack branching time (the maximum propagation velocity occurs at 36 μs).

In contrast, the XFEM + Meshfree model predicts significantly higher crack propagation velocity peaks than the other models,

Table 1
Comparison of the proposed method with alternative numerical approaches.

	Crack propagation onset time (μs)	Crack branching time (μs)	Maximum propagation speed (m/s)
Present	*10.3*	*30.5*	*1271*
PF-CZM	*12.5 (+21.36 %)*	*27.7 (+10.11 %)*	*1242 (−2.28 %)*
Peridynamics	*9 (−12.62 %)*	*36 (+18 %)*	*1349 (+6.14 %)*
XFEM + Meshfree	*7.9 (−23.30 %)*	*31 (+1.61 %)*	*2082 (+63.81 %)*
FEM-CZM	*11.2 (+8.74 %)*	*29.6 (−2.95 %)*	*1337 (5.19 %)*

including the proposed approach. It is important to clarify that Fig. 10 does not report velocities exceeding $0.6c_R$ because the branching criterion acting on the crack propagation velocity stops the analysis when this threshold is reached. However, detailed analyses conducted without activating the velocity criterion (not reported here for brevity) demonstrated that the maximum velocity predicted by the proposed method is only slightly higher than $0.6c_R$, never reaching speeds comparable to c_R.

This observation suggests that the velocity-based branching criterion has a minimal impact on the calibrated values of K_{IA} and m. Additionally, the close correlation between the crack tip velocity time history predicted by the proposed method and other numerical models (both before and after crack branching), which do not require an explicit curve for dynamic crack growth toughness, is noteworthy. This strong alignment suggests that the calibrated values for K_{IA} and m are likely to be very close to those of the actual material, thus demonstrating the effectiveness of the calibration procedure described in Section 4.1.1.

Table 1 reports detailed data regarding the crack propagation onset, branching times, and peak propagation velocity before branching, facilitating a detailed comparison between the proposed method and other methodologies. As one can see, despite some variations in crack initiation time, the crack branching time predicted by the proposed method closely matches the results obtained with PF-CZM, XFEM + Meshfree, and FEM-CZM methods, with a maximum difference of approximately 10.11 %. This agreement is particularly significant considering the diverse nature of the methodologies. Indeed, PF-CZM and FEM-CZM methods handle crack branching internally without an explicit criterion, whereas XFEM + Meshfree employs an external criterion distinct from the velocity criterion utilized in our proposed method.

Fig. 12 compares the final crack paths predicted by the proposed method and those obtained through other numerical procedures. Notably, the results of the proposed method and even those of other approaches differ considerably from those gained by Agwai et al. [49]. This discrepancy can be mainly attributed to the delayed branching time predicted in that particular simulation, as highlighted in Table 1.

In contrast, the proposed method shows strong agreement with the predictions of Nguyen and Wu [54], Hirmand and Papoulia [86], and Menouillard and Belytschko [82], despite some minor discrepancies in the results. These slight variations can likely be traced to differences in crack propagation velocities, as shown in the time histories presented in Fig. 10. In the proposed model, the propagation angle is influenced by the DSIFs, which depend on the velocity of the crack tip. Thus, even small differences in the velocity history can lead to variations in the crack path.

This hypothesis is supported by two main observations. First, none of the numerical methods produces identical crack velocity profiles: the predicted curves exhibit local fluctuations and phase shifts, which can significantly influence the overall crack evolution. In this context, the velocity profile obtained with the proposed method closely matches those reported by Nguyen and Wu [54] and Hirmand and Papoulia [86], which is directly reflected in the strong similarity of their predicted crack paths. Second, the method proposed by Menouillard and Belytschko [82] consistently predicts higher crack tip velocities. This difference in velocity results in a more pronounced deviation in the resulting crack trajectory after the crack branching time compared to the other models.

4.1.3. *Parametric study on the crack branching angle*

Although various criteria exist in the literature for defining the crack branching angle [3,75,79] (*i.e.*, criteria proving the angle subtended by the branching cracks soon after branching with respect to the original crack plane), the present results initially assumed $\alpha_B = 45^\circ$ as a trial value for early evaluation. Despite being a trial value, the resulting crack path is in good agreement with those obtained using other numerical methodologies, indicating the robustness of the proposed approach.

This consistency is primarily attributed to the MM technique, which dynamically adjusts the orientation of the branching segments. This ensures that results are not strictly dependent on the initially assigned branching angle. Indeed, these segments primarily act as branching segments in the analysis steps after the branching time, allowing their orientation to be progressively corrected to the effective direction determined by the maximum circumferential stress criterion.

To further highlight this capability of the MM technique, a parametric study has been conducted to assess the influence of the initial branching on the reliability of the proposed method. Specifically, the fracture behavior of the plate has been investigated by

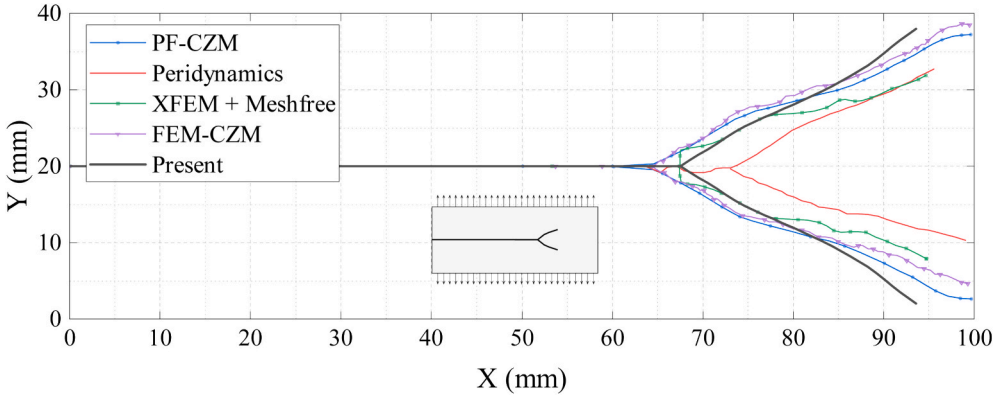


Fig. 12. A rectangular plate with a horizontal initial notch under tension. Comparison in terms of the crack path between the proposed model and the prediction gained by.

considering the initial branching angles of $\alpha_B = 30^\circ$ and $\alpha_B = 60^\circ$, in addition to $\alpha_B = 45^\circ$ trial value. Note that $\alpha_B = 60^\circ$ represents an extreme case not supported by experimental evidence, which typically reports values for the branching angles within the range of 10° to 45°. Nevertheless, such value has been considered to ensure a comprehensive parametric study and, additionally, to assess the robustness of the proposed method in simulating crack branching over a broad range of angles. Furthermore, $\alpha_B = 60^\circ$ corresponds to the theoretical value suggested by Yoffe [3]. Fig. 13 compares the final crack paths obtained by considering the initial values for branching angles of 30°, 45°, and 60°. Additionally, Fig. 14 illustrates snapshots of the computational mesh for the three cases under investigation immediately after the branching time.

The results of this parametric study clearly highlight that the crack paths remain largely identical across all tested initial angles. Only minor deviations are observed in the initial stage, directly attributable to the differences in the starting configurations, as can be observed from the zoomed views in Fig. 13. Consequently, these results unequivocally demonstrate that the initial value of the branching angle does not affect the reliability of the numerical solution predicted by the proposed method.

To further emphasize the key role played by the moving mesh, Fig. 15-a and b illustrate snapshots of the computational mesh for the numerical analysis with $\alpha_B = 60^\circ$ at $t = 30.5$ μs and $t = 31.5$ μs, respectively. These snapshots clearly demonstrate the dynamic adaptation of the computational mesh and, critically, the rapid adjustment of the initially assigned branching angle within just 1 μs after branching occurs. This visual evidence further supports the robustness of the Moving Mesh technique since it provides the significant benefit of intrinsically handling the crack branching condition within the simulation framework. This inherent capability obviates the need for an additional external criterion to evaluate crack branching, thus streamlining the modeling process and enhancing the method's predictive power.

4.2. A square plate with a horizontal initial notch under biaxial loading

Fig. 16 shows a Homalite-100 square plate with a side length of $L = 254$ mm and a thickness of $B = 16.4$ mm, having a central horizontal notch of length $a = 50$ mm. The plate is subjected to biaxial tensile loads σ_x and σ_y acting along the horizontal and vertical boundaries, respectively. Such loads are suddenly applied at $t = 0$ and then held constant over time.

Hawong et al. [88] conducted experiments to investigate the fracture behavior of this plate under different load ratios $\beta = \sigma_x/\sigma_y$, ranging from 0 to 3.7. For the specific case of $\beta = 0.33$, their experimental results indicate that both crack tips initially propagate horizontally towards the vertical boundaries of the plate, thereby extending the pre-existing notch. Before crack branching occurs, the right and left cracks grow by approximately 9 mm and 24 mm, respectively. In this context, Hawong et al. [88] report the evolution of mode-I DSIF of both crack tips as a function of crack extension. They observed that crack branching events occur when the mode-I DSIF reaches the critical stress intensity factor for branching (K_B) of the material. After branching, the crack tips split into trajectories inclined at approximately 20°–22° with respect to their preceding crack path and continue propagating. As the cracks move forward, the propagation angle with respect to the horizontal gradually decreases, leading to trajectories that progressively align horizontally.

To analyze the fracture behavior of the plate through the proposed model, the computational mesh shown in Fig. 17 has been employed. This mesh consists of 1324 triangular elements, finally arranged around the crack tip region, where four equally spaced elements discretize the stretching segments. Due to the relatively small thickness (16.4 mm) compared to the in-plane dimensions (254 mm x 254 mm), a plane stress formulation has been adopted in the present model. Additionally, results were recorded at a maximum time interval of 0.5 μs for data output and visualization purposes.

4.2.1. Calibration of the dynamic crack growth toughness curve of the material

Hawong et al. [88] reported the Young's modulus, Poisson's ratio, and critical stress intensity factor for branching for the Homalite-

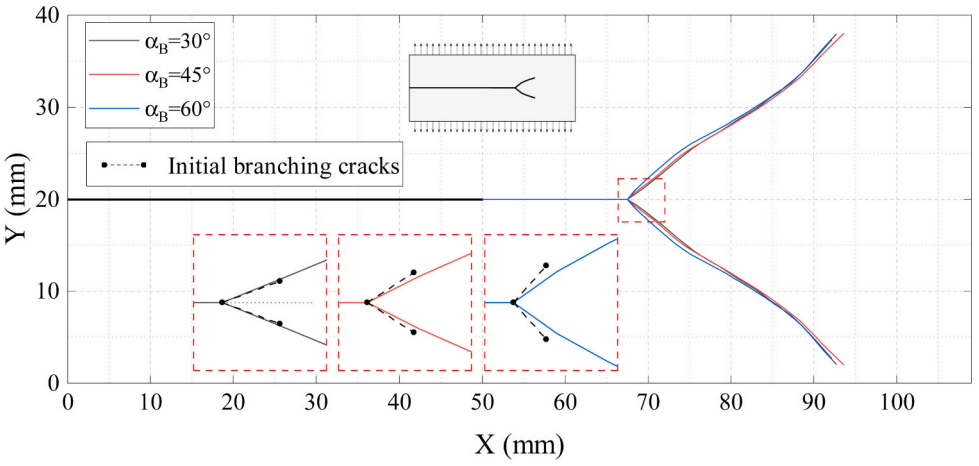


Fig. 13. A rectangular plate with a horizontal initial notch under tension. A comparison of final crack paths as a function of variable crack branching angles.

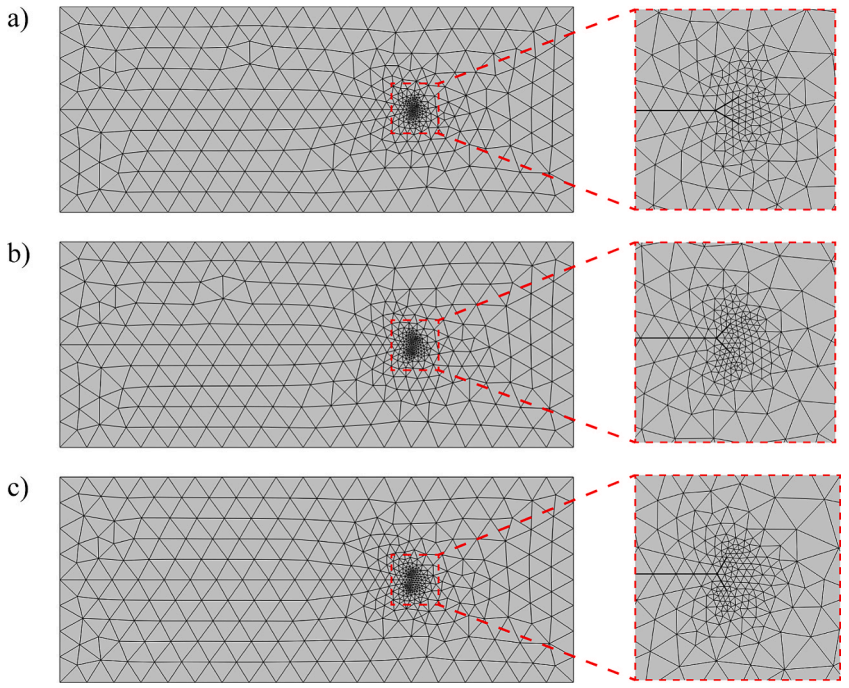


Fig. 14. A rectangular plate with a horizontal initial notch under tension. Computational mesh configuration immediately after crack branching time for different crack branching angles: (a) $\alpha_B = 30°$; (b) $\alpha_B = 45°$; (c) $\alpha_B = 60°$.

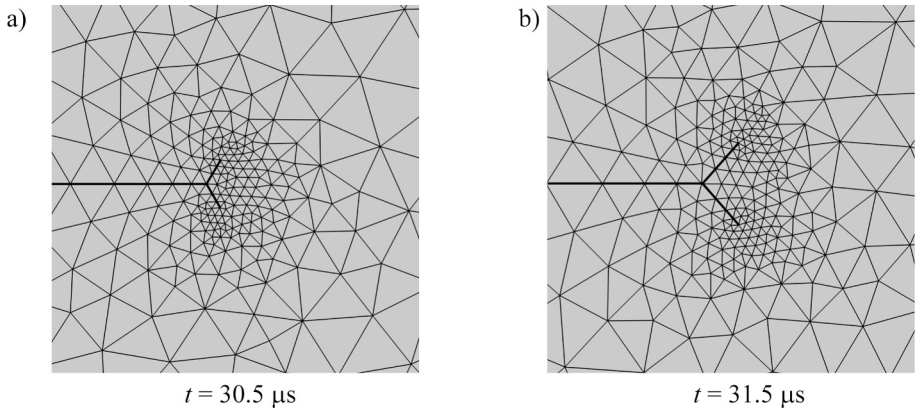


Fig. 15. A rectangular plate with a horizontal initial notch under tension. The evolution of the computational mesh during the steps immediately after the branching time for the case of $\alpha_B = 60$: (a) t = 30.5 μs; (b) t = 31.5 μs.

100 material as $E = 4.27$ GPa, $\nu = 0.357$, and $K_B = 200$ kPa m$^{1/2}$, respectively. Unfortunately, their work did not provide additional material properties like mass density (ρ), fracture toughness (K_{Id}), or crack arrest toughness (K_{IA}). Furthermore, the proposed model requires suitable values for the dimensionless shape parameter (m) and the limiting crack propagation speed (V_L) to fully define the dynamic crack growth toughness of the material, as described by Eq. (26).

To address these missing parameters, some properties have been consistently defined using the available literature data. In particular, $V_L = c_R = 1045$ m/s, $\rho = 1230$ kg/m^3, and $K_{Id} = K_B/2$ [89,90] were assumed. Note that the assumption for K_{Id} aligns with the main experimental observations on crack branching, which typically show the critical fracture toughness for branching to be two times larger than the fracture toughness of the material.

For the remaining unknown parameters, K_{IA} and m, a calibration analysis was performed. Similar to Section 4.1.1, acceptable ranges for these parameters were established as $0.5K_{Id} \leq K_{IA} \leq 0.9K_{Id}$ and $2 \leq m \leq 5$, respectively.

This calibration utilized the experimental curve reported in [88], which shows the evolution of the Mode-I DSIF as a function of crack extension (a) for the case $\beta = 0.33$ and $\sigma_x = 7.6$ MPa. Consistent with this experimental data, the Stress Intensity Factor criterion (see Eq. (28)) is adopted to determine the crack branching time.

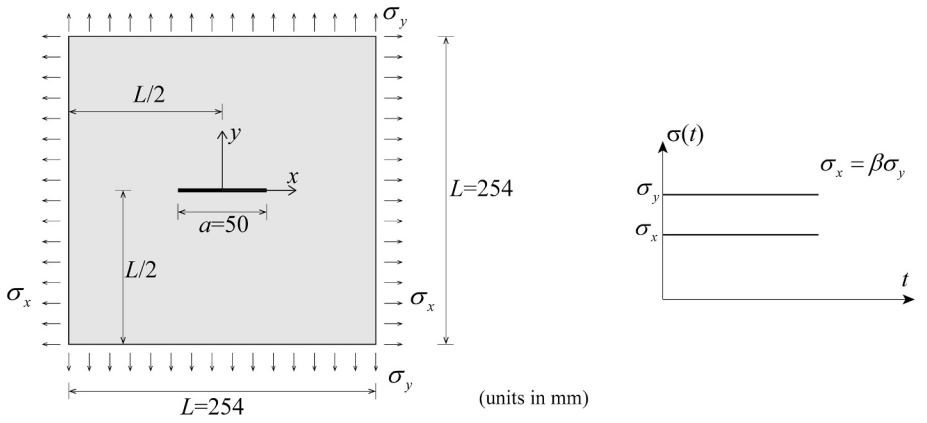


Fig. 16. A square plate with a horizontal initial notch under biaxial loading. Geometry and boundary conditions.

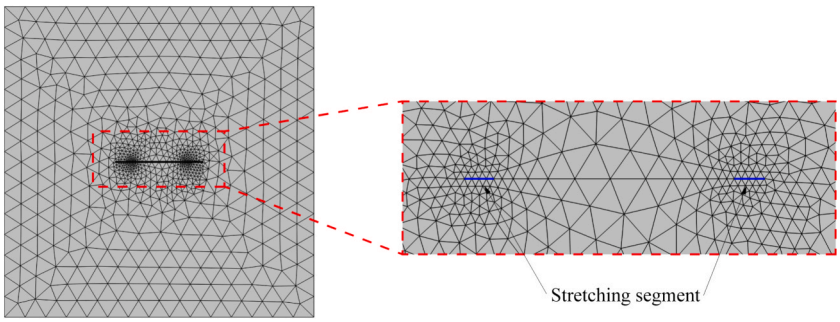


Fig. 17. A square plate with a horizontal initial notch under biaxial loading. Initial mesh configuration.

Initially, numerical simulations were performed using dynamic crack growth curves for the material defined by two sets of extreme values: $K_{IA} = 0.5K_{Id}$ and $m = 5$ (curve c_1 in Fig. 18-a), and $K_{IA} = 0.9K_{Id}$ and $m = 2$ (referred to as curve c_2 in Fig. 18-a). The resulting a vs DSIF graphs (Fig. 18-b) differed significantly from the experimental curve, leading to the exclusion of these curves from further consideration.

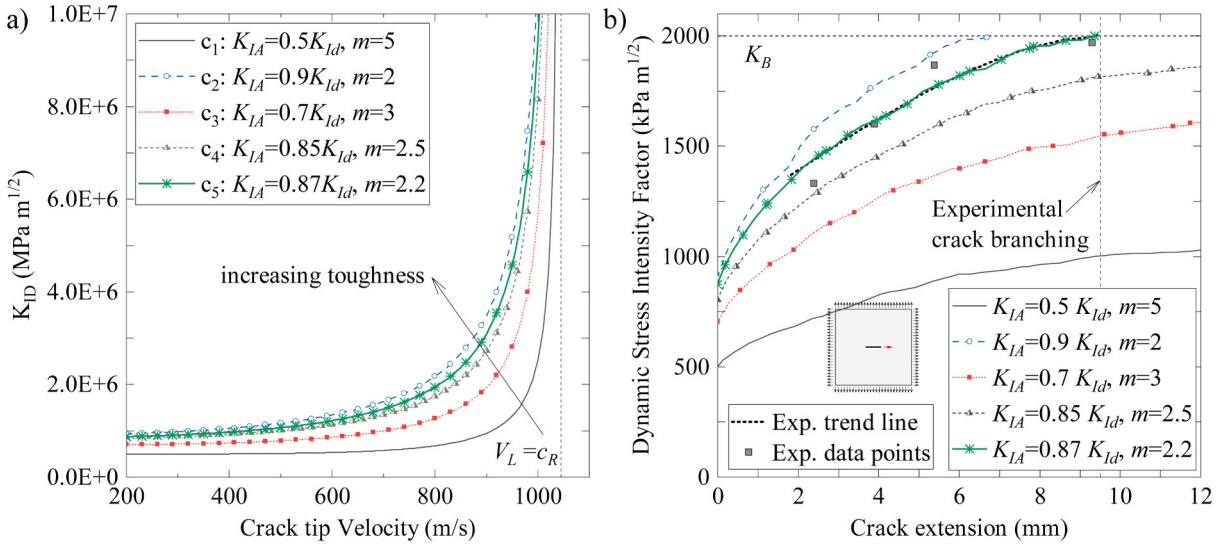


Fig. 18. A square plate with a horizontal initial notch under biaxial loading. A comparison in terms of mode-I dynamic stress intensity factor between the proposed method and experimental results for the time interval comprised between the beginning of the simulation and the crack branching time.

Next, an intermediate curve (c_3 in Fig. 18-a) was defined by assuming $K_{IA} = 0.7K_{Id}$ and $m = 3$. While the numerical results associated with curve c_3 still differed from the experimental one, they indicated the need for a curve with higher toughness. Consequently, a novel K_{ID} curve (c_4 in Fig. 18-a) was defined by selecting $K_{IA} = 0.85K_{Id}$ and $m = 2.5$. These parameters considerably improved the convergence of the numerical Mode-I DSIFs to the experimental curve. However, additional fine-tuning was necessary. Both parameters were further refined to $K_{IA} = 0.87K_{Id}$ and $m = 2.2$, corresponding to curve c_5 in Fig. 18-a, which lies between curves c_3 and c_4. This final configuration provided the best agreement with the experimental mode-I DSIF curve, as shown in Fig. 18-b. Additionally, it is important to note that these values implicitly capture the experimental crack extension, where crack branching occurs. Consequently, the calibrated values of K_{IA} and m were adopted for all subsequent numerical simulations. These values, which are material constants, are likely to be very close to the effective material values.

4.2.2. *Validation of the proposed approach*

Numerical predictions from the proposed approach were compared with results from other well-established numerical methods. Specifically, comparisons were made with the peridynamic model developed by Lee and Hong [90] (Peridynamic) and the Smoothed Particle Hydrodynamics (SPH) variants (Eulerian SPH, or ESPH, and Total Lagrangian SPH, or TLSPH) proposed by Islam et al. [75] (SLP and TLSP).

Fig. 19-a and Fig. 19-b present the time histories of the coordinates (referenced to the system in Fig. 16) and the cumulative crack extension of the right-upper crack tip, respectively. Fig. 19-c depicts the final crack path.

The results indicate that predictions from the proposed method exhibit overall good agreement with those of other numerical approaches. In particular, the time history of the x-coordinate and the cumulative crack length align with the predictions of other authors. In contrast, the time history of the y-coordinate shows larger values compared to those from other methodologies. This difference can likely be attributed to the fact that the proposed method predicts a crack branching time of $t_B = 76.5$ μs, which is approximately 20 % smaller than that predicted by other methodologies. The resulting delay in predicting the crack branching time by others results in a crack extension associated with branching that is roughly 57 % greater than the experimental data, which measures approximately 9.5 mm (as shown in Fig. 18-b). In contrast, the crack extension associated with crack branching predicted by the proposed method agrees with the experimental data (Fig. 18-b).

As a result, the post-branching crack trajectories predicted by the Preidynamic and TLSPH models develop later than those

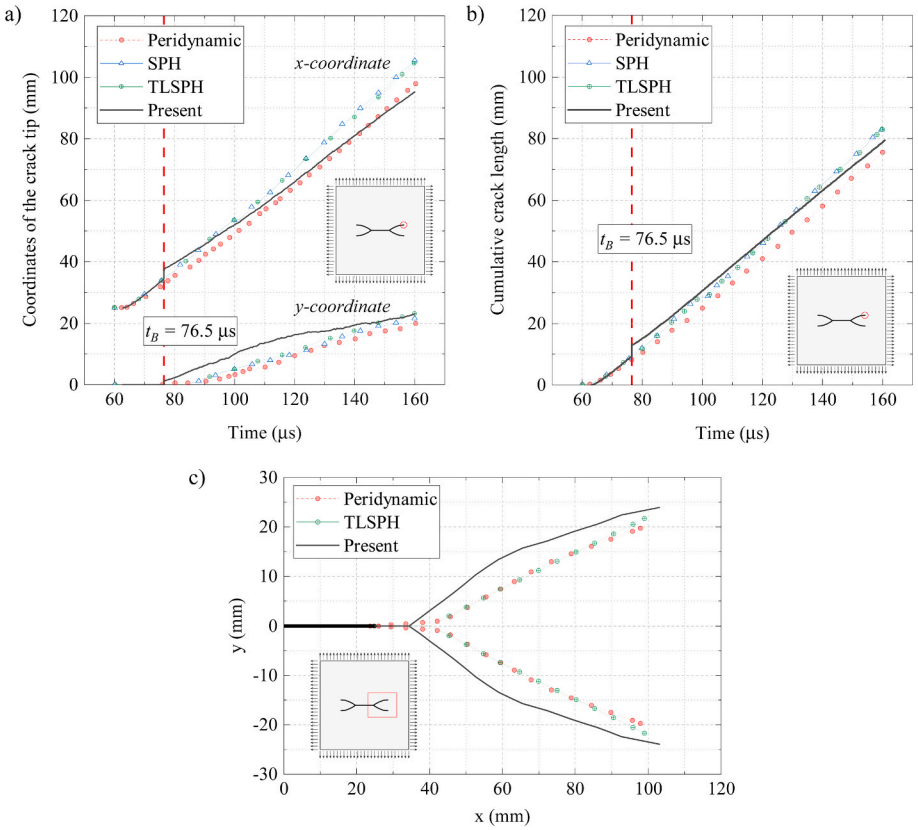


Fig. 19. A square plate with a horizontal initial notch under biaxial loading. A comparison between the predictions of the proposed models and the numerical results of Lee and Hong [90] and Islam et al. [89]. (a) Time history of the coordinates of the right crack tip; (b) Time history of the crack length of the right crack; (c) Final crack paths for the right crack.

predicted by the proposed method. These discrepancies in crack extension values are likely attributable to the specific constitutive models or inherent scale effects employed by each different methodology.

Nevertheless, it is notable that the inclination of trajectories relative to the horizontal predicted by the proposed method is similar to that of the Peridynamic and TLSPH methods. More generally, the overall qualitative shape and general direction of the branched crack paths are comparable. This collectively confirms the proposed model's ability to capture the phenomenon of dynamic crack branching accurately.

Note that the predicted time histories of the x and y coordinates (Fig. 19-a) show discontinuities at the moment of crack branching. These discontinuities occur because, at the branching instant, the proposed method adds two additional segments, each of length $l_B = 3.5$ mm, into the geometry to represent the splitting of the primary crack. These segments are inclined at approximately 22° with respect to the horizontal, a value close to that observed in experimental tests [88]. This geometric insertion inevitably introduces discrete steps in the time histories of the crack tip coordinates and cumulative crack length. However, it is crucial to emphasize that these discontinuities arise solely from the explicit geometric update made to represent the new crack branches and do not indicate numerical instability or artifacts.

Regarding the impact of these discontinuities, the resulting deviations are not excessive. In the time histories of the x-coordinate (Fig. 19-a, top) and the crack length (Fig. 19-b), the initial step is noticeable, but the subsequent trend of the predicted curve generally follows the trajectories of the SPH (and TLSPH) and Peridynamic methods. In contrast, the discontinuity is less evident in the time history of the y-coordinate of the crack tip (Fig. 19-a, bottom) due to the relatively small value of the branching angle (*i.e.*, $\alpha_B = 22°$).

Fig. 20 illustrates the configuration of the computational mesh at times $t = 76.5$ μs, $t = 120$ μs, and $t = 160$ μs. These results demonstrate how the proposed method adapts the computational mesh during the simulation. In particular, the refined zones around the crack tip advance with the crack fronts while the boundaries of the crack face become progressively coarser. In this context, regions far from the center of the plate maintain coarse meshes, significantly improving computational efficiency compared to traditional methods.

Fig. 21 compares the curving angle of the right crack obtained by the proposed method with that of the Peridynamic model developed by Lee and Hong [90]. This comparison represents an interesting result, as it highlights the effectiveness of the moving mesh technique in accurately managing the propagation direction, especially during the post-branching phase. Specifically, Fig. 21-a shows the time histories of the curving angle of the upper crack tip, while Fig. 21-b illustrates snapshots of the mesh configuration at the time instances indicated in Fig. 21-a with Roman numerals. In particular, the latter Figure illustrates the comparison between the proposed method and the peridynamic model developed by Lee and Hong [76] in terms of the tangent lines at the crack faces.

The results show a sudden drop in the time history of the curving angle predicted by the proposed method at $t = 76.5$ μs (*i.e.*, the moment of crack branching), reflecting an immediate increase in the angle to 22°. This change is due to the modeling strategy used to simulate the splitting of a primary propagating crack (described in Section 3), which involves inserting two additional segments representing the branching cracks within the geometry. These segments primarily serve as stretching segments for the mesh, establishing the initial geometric configuration for the newly formed branches. Importantly, their initial orientation of 22° is a numerical setup rather than a fixed physical assumption.

After branching, the curving angle exhibits an oscillatory behavior, initially increasing between 80 μs and 90 μs, decreasing up to 150 μs, and then rising again until the end of the simulation. During this phase, the Moving Mesh technique, guided by the established fracture criterion, continuously adjusts the orientation of these newly formed crack segments. This mechanism quickly re-aligns the crack path with the physically driven direction. Notably, the proposed methodology aligns well with the results obtained by Lee and Hong [90], starting from $t = 120$ μs. In particular, the snapshot at $t = 120$ μs (Fig. 21-b) confirms that both methods predict the same propagation direction.

These results clearly demonstrate the robustness of the proposed method. Despite the numerical initialization of the branched crack paths with an initial 22° angle, the method proves to be insensitive to this initial choice. The adaptive nature of the moving mesh ensures that the computational mesh effectively adjusts over time, progressively reducing the initial numerical slope to values consistent with other numerical predictions and the underlying physical criteria. This aspect verifies the reliability of the solution

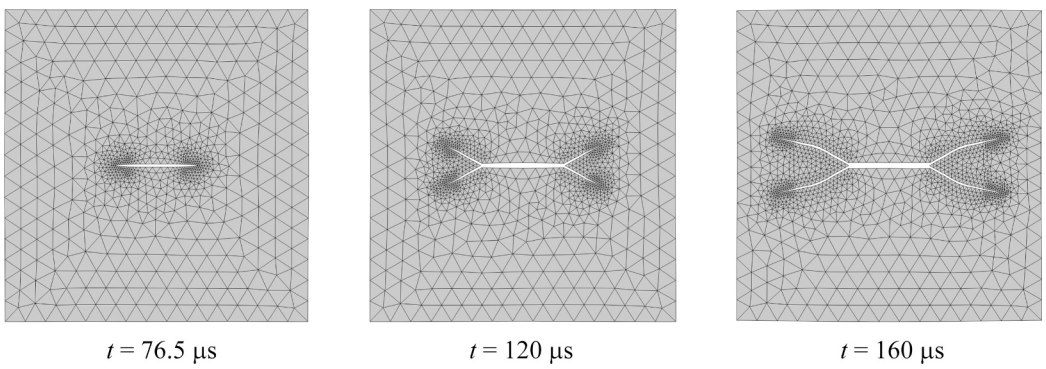


Fig. 20. A square plate with a horizontal initial notch under biaxial loading. The evolution of the computational mesh is due to the crack propagation process.

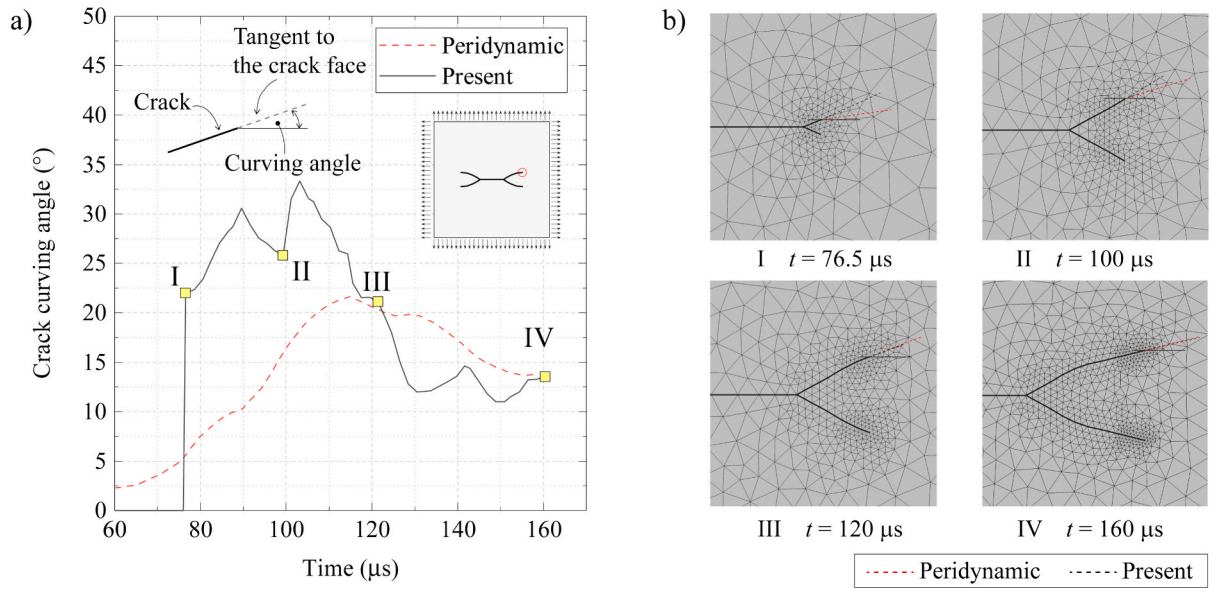


Fig. 21. A square plate with a horizontal initial notch under biaxial loading. A comparison between the prediction of the proposed method and the results achieved by Lee and Hong [90] in terms of (a) the time history of the crack curving angle; (b) Snapshots of the mesh configurations of the right side of the plate relative to the time instance marked by Roman numerals.

during the post-branching phase, regardless of the initial numerical setup of the branched crack segments.

4.3. An edge-cracked plate impacted by a projectile

This section aims to analyze the dynamic fracture behavior of a plate impacted by a projectile based on the experiment developed by Kalthoff and Winkler [91].

Fig. 22-a shows a square plate of side $L = 100$ mm with a horizontal notch of length $a = 50$ mm, positioned at a distance $H_a = 25$ mm from the bottom edge. The Young's modulus, Poisson's ratio, mass density, and dynamic crack initiation toughness are equal to $E = 190$ GPa, $\nu = 0.3$, $\rho = 8000$ kg/m^3, and $K_{Id} = 68$ MPa m$^{1/2}$, respectively. Correspondingly, the dilatational wave speed, the shear wave speed, and the Rayleigh wave speed are $c_D = 5108$ m/s, $c_S = 3022$ m/s, and $c_R = 2799$ m/s, respectively.

The projectile impacts the plate on the left boundary, between the bottom edge and the notch, causing the initial crack front to advance. According to experimental observations [91], the crack advances along a trajectory inclined at $\alpha = 70°$ with respect to the horizontal, without branching events.

This specific case has been extensively simulated numerically using various methods (see, for instance, [33,83,86]), as it represents a key benchmark case for mixed-mode dynamic crack propagation. Numerically, the impact is reproduced by imposing a constant velocity $v_0 = 16.5$ m/s on the impacted boundary. Literature results consistently reproduce the experimental crack path and provide reliable time histories of the crack propagation velocity.

Beyond the experimental scenario, several authors have numerically investigated the fracture behavior of the plate for impact velocities higher than 16.5 m/s. These studies indicated that higher impact velocities can lead to multiple branching events inside the plate during crack growth.

To assess the ability of the proposed methodology to reproduce dynamic crack propagation events involving branching under mixed-mode conditions, the fracture behavior of the plate has been examined considering an impact velocity $v_0 = 40$ m/s. For these simulations, the computational mesh depicted in Fig. 22-b serves as the initial discretization of the plate geometry. It consists of 802 plane stress triangular elements arranged finely around the crack tip region and coarser elsewhere. Notably, for this case, an additional segment of 3.5 mm inclined at 15° with respect to the horizontal was inserted beyond the initial pre-crack (see the zoomed view of Fig. 22-b). This segment serves as a stretching segment, thereby preserving the initial configuration of the pre-crack and it is discretized with three elements of variable length.

Accurate estimation of the fracture behavior of the plate through the proposed method necessitates a proper dynamic crack growth toughness curve (K_{ID}) for the material. According to Eq. (26), K_{ID} requires the crack arrest toughness (K_{IA}), a suitable dimensionless shape parameter (m), and the limiting propagation speed (V_L). This latter parameter is typically set equal to the Rayleigh wave speed (c_R). Ideally, K_{IA} and m should be determined through experimental tests. Unfortunately, these parameters are unavailable for the specific material under investigation. Consequently, they were determined through a calibration analysis, employing the same approach as in Sections 4.1.1 and 4.2.1.

For this specific calibration, the reference data consisted of the experimental crack trajectory and the numerical time history of crack propagation velocity from the literature, specifically those reported for the case of an impact velocity $v_0 = 16.5$ m/s. The

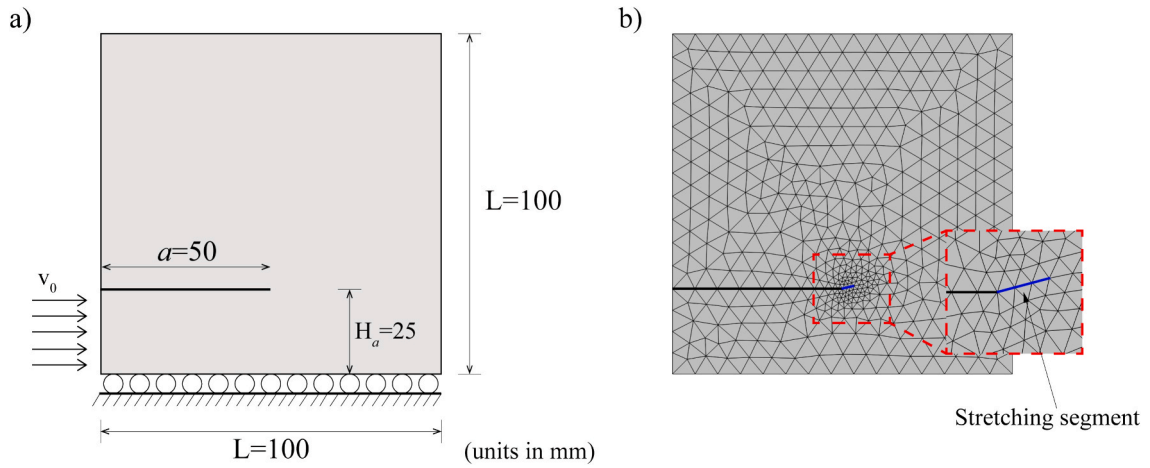


Fig. 22. An edge-cracked plate impacted by a projectile. (a) A depiction of the geometry and boundary conditions. (b) Initial mesh configuration adopted in numerical simulations.

maximum time step for data output was set equal to 1 μs, which ensures a good dataset for post-processing.

The complete calibration results, which are not reported here for brevity, identified the optimal values of $K_{IA} = 0.88\, K_{Id}$, $m = 3$, and $V_L = c_R = 2799$ m/s for accurately replicating the experimental crack path and the numerical time history of crack propagation velocity. In particular, the calibration was based on the numerical results from Tran and Bui [92], Borden et al. [46], and Braun and Fernández-Sáez [93], who simulated the fracture behavior of the plate using a nonlocal gradient damage model, a phase field model, and a discrete model, respectively.

Fig. 23-a shows the time history of the crack tip velocity, including comparisons with results from other numerical methods. Fig. 23-b illustrates the configuration of the computational mesh at the end of the simulation (note that the observed mesh overlap is a visualization artifact due to the scale effects of the mesh, not a physical phenomenon).

The results demonstrate that the time history curve obtained using the calibrated parameters aligns closely with those from other numerical methodologies. There is a notable agreement in terms of maximum propagation velocity and the timing of deceleration. Additionally, the crack path obtained corresponds well with experimental evidence, following a trajectory inclined at approximately 70° with respect to the horizontal.

The achieved values for K_{IA} and m were used to analyze the fracture behavior of the plate subjected to an impact velocity of $v_0 = 40$ m/s, which represents a different loading scenario compared to that used for the calibration analysis.

In this scenario, the velocity criterion was employed to assess crack branching conditions, assuming a critical velocity of $0.6c_R$. This critical value has been consistently identified in studies by numerous authors who have investigated the dynamic fracture behavior of

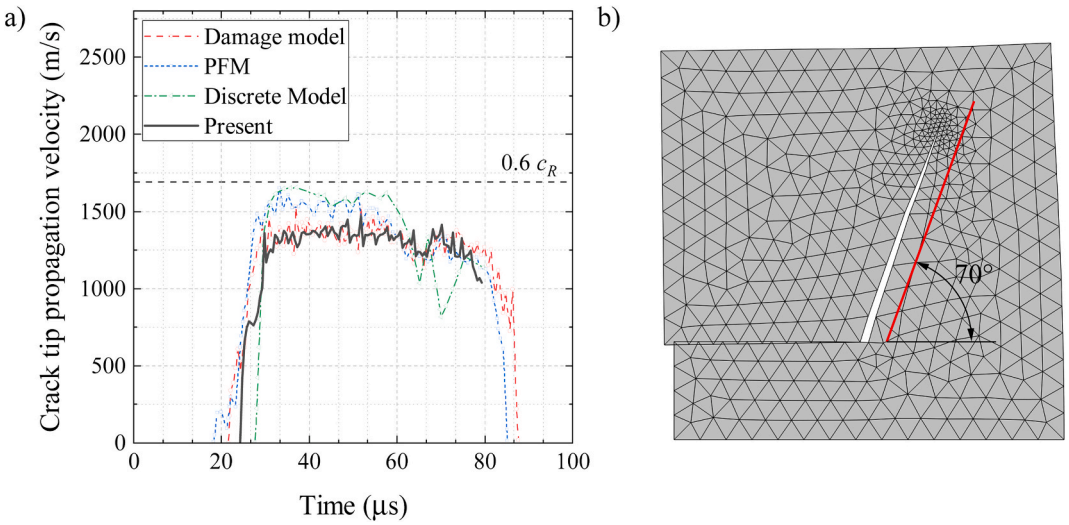


Fig. 23. An edge-cracked plate impacted by a projectile: calibration analysis. (a) A comparison in terms of the time history of a crack propagation velocity between the proposed approach and predictions of Tran and Bui [92] (Nonlocal gradient damage model), Borden et al. [46] (PFM), and Braun and Fernández-Sáez [93] (Discrete model); (b) Final crack path.

plates numerically (see, for instance, [92,94]).

Fig. 24 presents the time history of the crack propagation velocity, while Fig. 25 illustrates snapshots of the computational mesh configuration at different time instances. The results show that a projectile impacting the plate at a velocity of 40 m/s initiates a dynamic crack propagation mechanism within the plate. This mechanism leads to the growth of the initial crack front, designated as Tip I (see labels in Fig. 24), alongside two distinct crack branching events.

Specifically, once the crack propagation mechanism begins, Tip I grows initially along a trajectory inclined at approximately 70° with respect to the horizontal. However, this inclination decreases slightly as the crack propagation progresses, thus marking a first difference from the experimental scenario (compare Fig. 23-b with Fig. 25-I). At $t_{B1} = 33$ μs, the propagation velocity reaches the threshold value ($0.6c_R$), thus triggering the first crack branching event. The proposed procedure halts the analysis and updates the geometry of the plate by inserting two additional segments that replicate the splitting of the crack front (Tip I). These segments have a length of $l_B = 3.5$ mm and are inclined at $\alpha_B = 30°$ with respect to the tangent direction of the crack surfaces at Tip I. This results in two new crack fronts: Tip II and Tip III.

Following the geometry update, the computational domain is meshed, and the analysis resumes. All crack fronts within the plate (namely, Tip II and Tip III) dynamically grow (see Fig. 25-II). In this context, Fig. 24 highlights that the crack propagation velocity of Tip II decreases progressively until $t = 52$ μs, thus marking a crack arrest event. In contrast, the propagation velocity of Tip III, after an initial decrement, suddenly increases up to the critical threshold of $0.6c_R$ at $t_{B2} = 43$ μs, leading to a secondary branching event. This results in the splitting of Tip III into two new crack fronts (Fig. 25-III), referred to as Tip IV and Tip V.

The newly formed crack fronts, Tip IV and Tip V, evolve with distinct velocities and trajectories. Specifically, the propagation velocity of Tip IV exhibits a decreasing trend. In contrast, Tip V initially grows at a velocity approximately equal to 1500 m/s up to $t =$ 60 μs, after which it also begins to decrease. The observed decrements in crack tip propagation velocity are likely to be attributable to the fact that crack fronts approach the edges of the plate.

The predictions of the proposed method regarding the crack path are compared with the results obtained by Hai et al. [95], who reproduced the crack propagation inside the plate due to an impacted projectile with a velocity of 40 m/s using an explicit phase field model integrating viscoelasticity and micro-viscosity (Fig. 26).

The proposed method yielded comparable results to those of Hai et al. [95], especially in terms of the locations of crack branching and the overall trajectories of the cracks.

However, a notable difference arises concerning an additional crack path predicted by the model of Hai et al. [95]. Their simulation demonstrates crack propagation starting from the bottom-right corner of the plate and extending toward the initial notch. It is important to emphasize that neither the experimental setup nor the numerical model had a pre-existing notch or damage at this location. This implies that the model of Hai et al. [95] naturally initiates a damage zone. Unfortunately, the proposed method does not allow the initiation of a damaged front from an initially intact region. Indeed, an initial notch is always required for the $M-$integral computation. The slight discrepancies observed between our results and those of Hai et al. may stem from this additional damage zone, although it has a minimal influence on the overall development of the internal fracture fronts.

Despite this limitation, the overall reliability of our proposed model for predicting dynamic crack propagation is confirmed.

Indeed, the good comparability of our results with those of Hai et al. [95], whose phase-field model reproduces fracture front evolution without requiring explicit branching criteria, reinforces this consistency and provides strong additional validation for our proposed method.

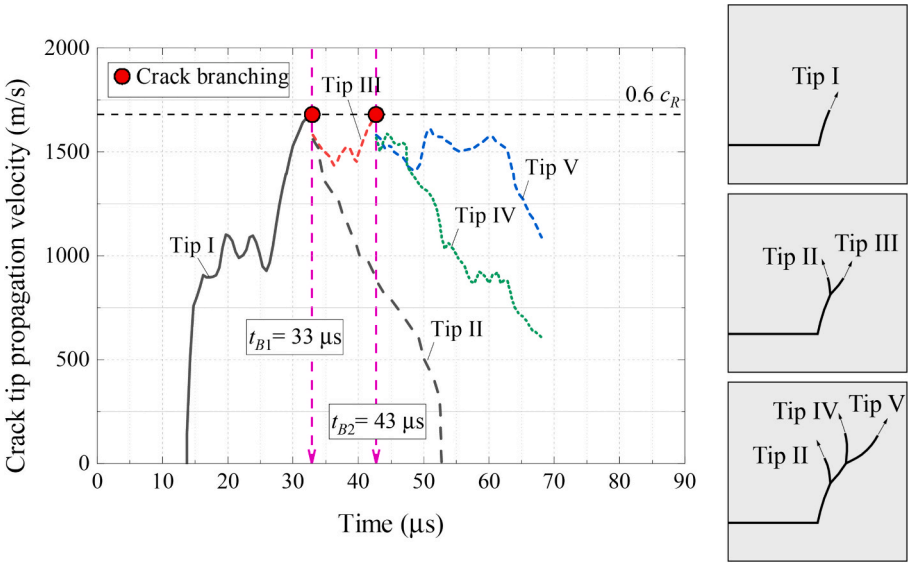


Fig. 24. An edge-cracked plate impacted by a projectile. Time history of the crack propagation velocity for the case with $v_0 = 40$ m/s.

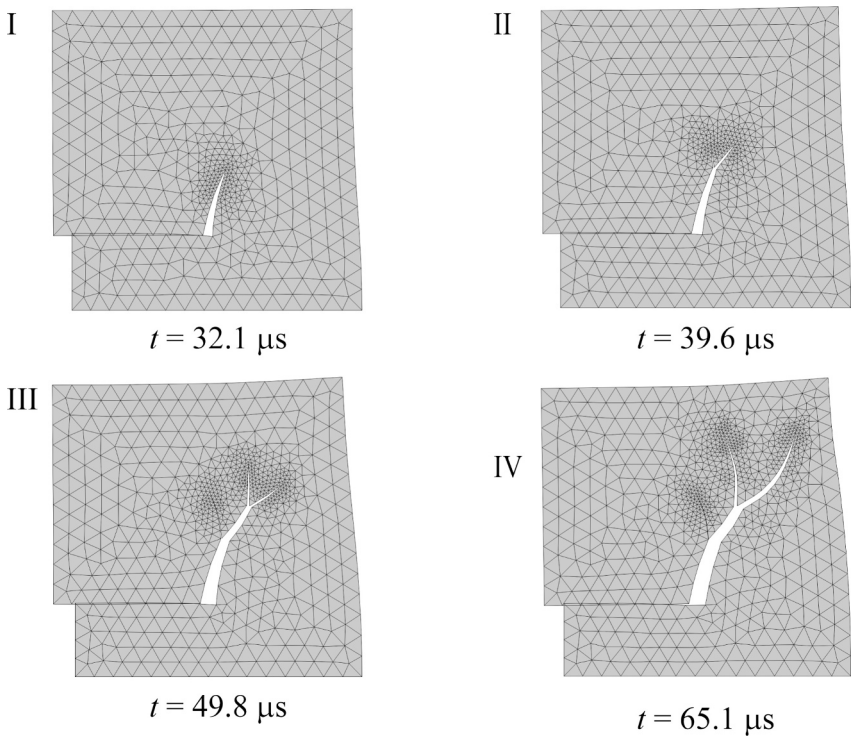


Fig. 25. An edge-cracked plate impacted by a projectile. Snapshots of the computational mesh configuration at different time instances associated with the case of study with impact velocity equal to 40 m/s.

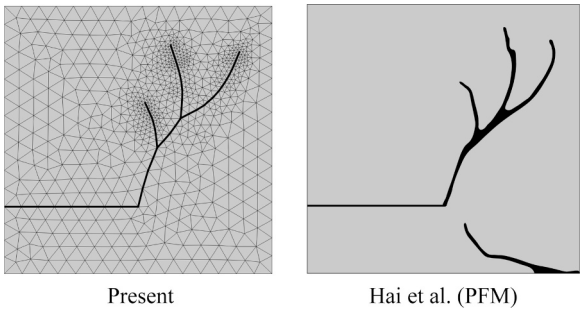


Fig. 26. An edge-cracked plate impacted by a projectile. Comparison in terms of the crack path between the proposed method and Hai et al. [95].

Finally, it is worth noting that the prediction of these complex branching mechanisms was obtained using values of K_{IA} and m for the material calibrated solely from a non-branching scenario, *i.e.,* the case of an impact velocity of 16.5 m/s. This aspect critically demonstrates the model's robust predictive capability across different fracture regimes and loading conditions. In particular, it reinforces that the calibrated parameters accurately represent the material's inherent dynamic fracture properties.

5. Conclusions

This work has presented an efficient and robust FE-based modeling strategy for simulating dynamic crack propagation and branching in brittle materials. The proposed modeling strategy leverages the Moving Mesh (MM) technique based on the Arbitrary Lagrangian–Eulerian (ALE) framework to reproduce growing cracks. More precisely, the computational mesh nodes associated with the crack tip move according to classic fracture criteria, thus reproducing the crack growth mechanism. In this context, the ALE formulation ensures that the finite elements deform with regularity during mesh motion, thereby preventing excessive distortions that can negatively impact numerical stability and accuracy.

The proposed modeling strategy addresses a well-known limitation of standard finite element methods, which, despite their simplicity and widespread use, are not inherently suited for tracking evolving discontinuities such as cracks. In particular, simulating dynamic crack propagation, especially crack branching, typically requires frequent remeshing and complex crack-tracking strategies,

which significantly increase computational costs and may reduce numerical stability.

The proposed approach overcomes these limitations by dynamically adapting the computational mesh to the evolving geometry through the MM technique, thus minimizing the need for remeshing actions. Therefore, the proposed approach enhances the capabilities of standard FEM while preserving its flexibility and generality for a wide range of fracture mechanics problems.

The reliability of the proposed method has been assessed through three benchmark cases by comparing the predicted results with experimental data and numerical simulations available in the literature. The outcomes demonstrate that the proposed strategy provides accurate predictions of crack trajectories, particularly in terms of the crack branching time.

Beyond the overall accuracy, the analysis of the results highlights specific strengths of the proposed modeling strategy. A key advantage lies in avoiding the imposition of predefined criteria for the branching angle at the moment of crack bifurcation. Indeed, the MM technique allows the branching direction to emerge naturally from the solution as it adapts to the stress field dictated by the maximum hoop stress criterion. This feature contrasts with traditional FE-based approaches, where the orientation of crack branches often needs to be explicitly defined. Therefore, the proposed method offers a more physically consistent and automated representation of crack evolution, especially in complex dynamic scenarios.

Despite the promising results, some limitations of the proposed method should be acknowledged. The first limitation concerns the selection of the fracture function used to assess the onset of crack branching. The current implementation allows users to choose between velocity-based and DSIFs-based criteria. However, the numerical predictions may be sensitive to this choice.

A second limitation lies in the need for several parameters to characterize the material's dynamic fracture toughness. These parameters are often not directly available from the literature and must be calibrated through dedicated simulations, which can be time-consuming. Specifically, in this work, a suitable calibration procedure was necessary to determine appropriate values for the crack arrest toughness (K_{IA}) and the dimensionless shape parameter (m) to define the curve of the dynamic crack growth toughness of the material. The results demonstrated that the calibration procedure used to determine these material constants yielded values closely resembling those that are likely to be the effective properties of the material. Particularly, the comprehensive analysis presented in Section 4.3 (Kalthoff-Winkler test) further substantiated the robustness and predictive capabilities of these calibrated parameters. In that particular case study, parameters calibrated from a non-branching scenario (16.5 m/s impact velocity) were successfully utilized to predict complex crack branching behavior under significantly different loading conditions (40 m/s impact velocity), confirming the model's accuracy across varied fracture regimes and reinforcing the reliability of these material constants.

Finally, a third limitation concerns the fact that the current form of the model cannot simulate micro-branching phenomena. This is due to the absence of an explicit criterion or mechanism specifically designed to trigger such instabilities. Capturing micro-branching would require additional modeling strategies, such as incorporating instability-driven criteria or extending the formulation to three dimensions.

Future developments will aim to reduce this dependency by formulating a fracture model that can be more directly parameterized from experimental data, enhancing both the physical accuracy and the general applicability of the approach. Furthermore, efforts will focus on exploring branching criteria inherent to the model to eliminate the reliance on the arbitrary selection of a specific fracture function. Meanwhile, the possibility of implementing proper fracture criteria to simulate micro-branching phenomena will be explored.

CRediT authorship contribution statement

Domenico Ammendolea: Writing – review & editing, Software, Methodology, Conceptualization. **Francesco Fabbrocino:** Supervision, Project administration, Funding acquisition, Conceptualization. **Lorenzo Leonetti:** Investigation, Formal analysis, Conceptualization. **Paolo Lonetti:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Arturo Pascuzzo:** Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors are unable or have chosen not to specify which data has been used.

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